Limits to Arbitrage in the Indian Equity Market

A BS project report submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Science in Economics

by

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Report Approval

This report entitled **Limits to Arbitrage in the Indian Equity Market** by **Prince Kumar**, Roll No. 23B3032, is approved for the degree of **Bachelor of Science in Economics** from the Indian Institute of Technology Bombay.

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Certificate

This is to certify that the report entitled **Limits to Arbitrage in the Indian Equity Market**, submitted by **Prince Kumar** (Roll No. 23B3032) to the Indian Institute of Technology Bombay for the award of the degree of **Bachelor of Science in Economics**, is a record of original, bona fide research work carried out under my supervision and guidance. The report has reached the standards fulfilling the requirements of the regulations related to the award of the degree.

The results contained in this report have not been submitted in part or in full to any other university or institute for the award of any degree or diploma, to the best of my knowledge.

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Declaration

I declare that this written submission represents my ideas in my own words and that, where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any idea, data, fact or source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have not been properly cited or from whom proper permission has not been taken when needed.

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Prince Kumar

Abstract

In theory, a share and a future on that share should be priced consistently, and arbitrage should close any gap. In practice costs, short-sale constraints, limited capital and position limits allow gaps to persist. I study how strongly these limits bind in Indian equities, where most of them change on known dates. A model of constrained arbitrage, in which India's F&O ban is a rule that lets every participant only reduce positions, gives predictions and the results that would contradict them; the tests were written down and frozen before any outcome was related to any treatment. The data are built from public exchange files for 2015 to 2026.

Bans bind hard on trading: futures volume falls by about two log points, and when the rule is enforced on intraday positions every futures trade closes positions. Correction of the futures-cash basis does not become one-sided in the way the model predicts: rich futures revert faster in a ban, mostly at the start of a spell. The basis falls by about six basis points when a ban begins, more when open interest was building fast, and there is no bunching of utilisation below the trigger. Neither result appears at false ban dates. The basis follows a wide, asymmetric band of about −35 to +24 basis points around each stock's median, and the full cost of a new cash-and-carry position lies in its outer tail. Measured against the old rule, the October 2025 reform cut ban days by about three quarters. Entry into the F&O segment speeds up the response of prices to market news but does not weaken drift after bad news. A constraint index built only from frictions predicts slower price adjustment across the whole market, also within stocks and after controlling for size.

BAN SPELLS, 2015 TO 2026	FUTURES VOLUME IN A BAN	BASIS WHEN A BAN STARTS
1,788	**−2.0 log points**	**−6.3 bp**
BAND AROUND THE MEDIAN	**BAN DAYS, NEW VS OLD RULE**	**DELAY AFTER F&O ENTRY**
−35 to +24 bp	**−73%**	**−0.11**

Ban spells: 2015 to October 2026. Volume: effect of a ban on log futures contracts traded. Basis: change at the start of a ban against matched controls. Band: pooled threshold estimates. Reform: ban days in the first year of the new rule against the old rule applied to the same stocks. Delay: change in price delay after entry into the F&O segment (median before entry 0.24).

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Abbreviations

ADDV	Average daily delivery (in shares), used in the position-limit formula from October 2025
AMFI	Association of Mutual Funds in India
BHAR	Buy-and-hold abnormal return
EAR	Earnings-announcement return (abnormal return over days -1 to $+1$)
F&O	Futures and options; the derivatives segment of the National Stock Exchange
FutEq	Futures-equivalent (delta-adjusted) open interest
IVOL	Idiosyncratic volatility
LSTAR / ESTAR	Logistic and exponential smooth-transition autoregressive models
MWPL	Market-wide position limit
NSE	National Stock Exchange of India
OI	Open interest
RD	Regression discontinuity
SEBI	Securities and Exchange Board of India
SETAR	Self-exciting threshold autoregressive model
SLB	Securities lending and borrowing
STT	Securities transaction tax
VWAP	Volume-weighted average price

Symbols

m	Futures-cash basis, $\ln F - \ln F^*$, in basis points
F, S, F^*	Futures price, stock price, cost-of-carry value of the future
$\mathrm{PV}(D)$	Present value of dividends going ex before expiry
r, τ	Three-month interbank rate; time to expiry in years
u	Relative demand pressure of outside traders toward futures
Φ	Combined depth of the cash and futures markets
a	Arbitrage position (units of cash-and-carry)
$\gamma = \alpha\sigma^2$	Risk aversion per unit of capital times the risk of the basis
c_+, c_-, c_u	Cost of a cash-and-carry position, of a reverse position, of unwinding
ω	Utilisation of the position limit, open interest divided by the MWPL
$\pi(\omega)$, ℓ	Probability of meeting a ban; extra cost of exiting in a ban
χ, λ	Share of blocked new demand that moves to cash; $\lambda = \chi\psi_F/\psi_S$
R	$(\Delta OI + V)/V$, zero when every futures trade closes positions
C	Composition of open interest before a ban: change in open interest over volume
U, L	Indicators of the basis lying in the stock's upper or lower tail

CHAPTER 1

Introduction

Why price gaps between a share and its future can persist, why India is a good place to measure the limits that allow them, how the work was organised to avoid fitting theory and data to each other, and what it finds.

1.1. THE QUESTION

When two ways of holding the same risk trade at different prices, an arbitrageur can buy the cheap one, sell the dear one and wait for the prices to meet. If arbitrage were free and unlimited, such gaps would never be large or long-lived. Shleifer and Vishny (1997) and Gromb and Vayanos (2010) explain why it is neither: arbitrage needs capital, it carries risk while the gap persists, it costs money to trade, and it often needs a short position that may be costly or not allowed. Each of these limits lets prices drift away from each other.

This thesis asks how strongly these limits bind in the Indian equity market, which of them matter, and how much mispricing they explain. India is a good place to ask. Almost every limit changes on known dates: the securities transaction tax (STT) on futures was raised three times between 2023 and 2026; only stocks in the futures and options (F&O) segment can be sold short or shorted through stock futures, and stocks enter and leave that segment every few months; the assets of arbitrage mutual funds grew several times after a change in the taxation of debt funds in 2023; and position limits are set by formula and were changed in October 2025.

India also has an unusual rule. When market-wide open interest in a stock's derivatives exceeds 95% of its market-wide position limit (MWPL), the stock enters an *F&O ban*: every participant, not only arbitrageurs, may only trade to reduce positions until open interest falls to 80%. Between 2015 and October 2026 this happened 1,788 times to 141 stocks. To my knowledge no study has estimated its effects. The ban is the sharpest lens on one limit, but it is one lens among several; the thesis studies the limits side by side. Figure 1.1 shows how the rule works for the stock that was banned most often.

1.2. APPROACH

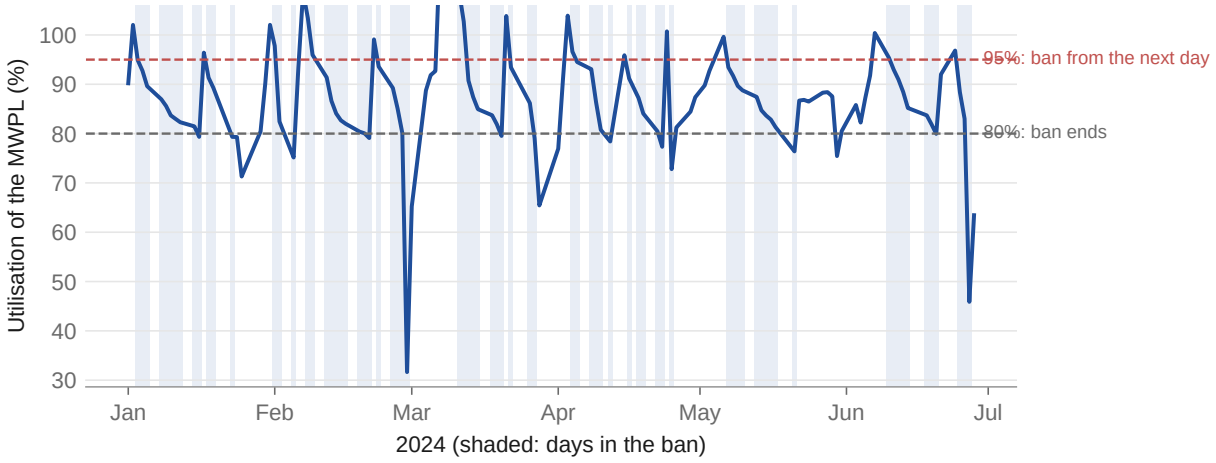
The work followed a fixed order, designed to prevent fitting the theory and the data to each other.

1. *Theory first.* One model covers all the limits; each enters as a parameter. The ban is modelled as the rule actually works. Every hypothesis is derived from the model with the

FIGURE 1.1

A stock enters the ban when open interest crosses 95% of its limit and leaves at 80%

Utilisation of the market-wide position limit for SAIL, January to June 2024; shaded days are ban days

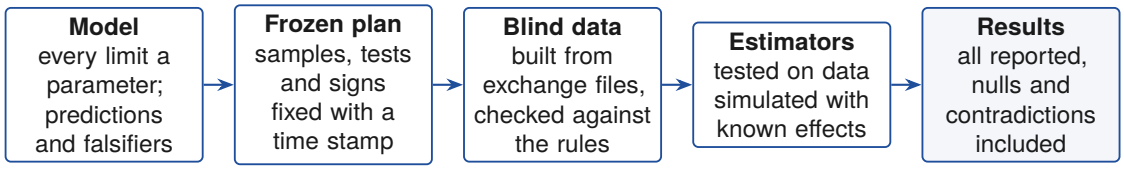


Source: NSE combined open interest files and securities-in-ban lists; author's calculations.

FIGURE 1.2

Theory and tests were fixed before any outcome was related to any treatment

The sequence of the project, from model to results



Every detail the plan left open was fixed in a dated log before the test that needed it (Appendix A).

result that would contradict it (Chapter 4).

2. *A frozen test plan.* Samples, variables, specifications and falsifiers were written down and frozen with a time-stamped record before any outcome was related to any treatment. Every detail the plan left open was fixed in a dated log before the test that needed it was run (Appendix A).
3. *Data from primary sources.* All data come from public exchange archives and are built with scripts that check every step against the exchange's own rules (Chapter 5).
4. *Estimators tested on simulated data* from the model before use.
5. *All results reported,* including those that contradict the model.

1.3. MAIN FINDINGS

1. **Bans bind on trading.** In a ban, futures trading falls by about two log points and the delivery share of cash trading rises. When the rule was enforced on peak intraday positions

(March 2020 to September 2025), the change in open interest equals minus the volume: every futures trade closed a long and a short. Under end-of-day enforcement before March 2020 this identity did not hold.

2. **Correction is not one-sided as predicted.** The model predicts that rich futures stop being corrected in a ban because nobody may open new cash-and-carry positions. The data show the opposite: rich futures revert faster in a ban, and cheap futures more slowly. The effect is concentrated in the first days of a spell and survives a correction for noise in daily prices.
3. **The basis falls when a ban starts.** Against stocks that came close to the trigger but did not cross it, the basis is about 6 basis points (bp) lower in the first days of a ban, more when open interest had been building fast, and it returns to its earlier level after about a week. Utilisation does not bunch below the trigger, and as utilisation rises the basis is more often rich.
4. **Costs, capital and the band.** The basis follows a wide, asymmetric band, about -35 to $+24$ bp around each stock's median. The upper part of the distribution rose around the 2024 and 2026 STT increases, but by more than the cost changes and mostly as a common shift. A new cash-and-carry round trip costs 27 to 31 bp, which places it in the outer tail of the basis; rolling an existing position costs 4 to 8 bp. Inflows into arbitrage funds are followed by a lower upper tail, but the evidence is weak. Measured against the old rule, the October 2025 changes cut the number of ban days by about three quarters, and the 2020 restoration of halved limits made a rich basis less common.
5. **Short-sale access and the cross-section.** After a stock enters the F&O segment its price responds faster to market-wide news; drift after bad news does not weaken. A constraint index built only from frictions, for every listed stock, predicts slower price adjustment, also after controlling for size; its link with drift after bad news is a size effect.

1.4. CONTRIBUTION

The thesis studies the F&O ban for the first time; models it as a reduce-only rule for every participant and shows that this makes correction one-sided in theory and the level effect ambiguous in sign; shows that changes in the position limit cannot serve as an instrument for ban status; estimates the no-arbitrage band of Indian stock futures over a long daily sample with threshold and smooth-transition models; and publishes a constraint index for all Indian stocks. The closest earlier work on Indian stock futures estimates bands from intraday data over short samples without the ban (Aggarwal, 2015; Shankar, Sankar and Kiran Kumar, 2019); Saurav, Srivastava and Agarwalla (2026) study the basis around earnings announcements and short-sale constraints.

1.5. OUTLINE

Chapter 2 reviews the related literature. Chapter 3 describes the Indian rules. Chapter 4 presents the model. Chapter 5 describes the data, the measures and the constraint index. Chapter 6 estimates the band and relates it to costs, capital and position limits. Chapter 7 studies the ban. Chapter 8 studies entry into and exit from the F&O segment and the cross-section of stocks. Chapter 9 concludes and discusses policy.

CHAPTER 2

Related literature

Four strands of work frame the thesis: the theory of why arbitrage is limited, the econometrics of the band within which a futures basis moves, the evidence on short-sale constraints and prices, and the evidence from India. The chapter ends with what is new here.

2.1. WHY ARBITRAGE IS LIMITED

Shleifer and Vishny (1997) start from the fact that arbitrage is done by specialists who invest other people's money. Investors judge them by past returns, so when a mispricing widens and the arbitrageur loses money, funds are withdrawn exactly when the opportunity is best. In their three-period model the price can fall by more than one for one with a deepening shock, because arbitrageurs are forced to sell into the fall. Two of their points matter here. Arbitrage capital is limited and depends on the state of the market, and specialised arbitrageurs avoid positions whose returns are volatile, so mispricing survives longest where idiosyncratic volatility is high.

De Long, Shleifer, Summers and Waldmann (1990) show that risk created by noise traders is enough to keep prices away from fundamentals, even when the fundamental value is certain. Arbitrageurs with short horizons must sell at a price that depends on future sentiment, and this risk deters them. The result needs the noise to be market-wide: misperceptions that are specific to one asset can be diversified away by arbitrageurs who hold many such positions, so the model gives a weaker reason to expect stock-level mispricing to rise with idiosyncratic volatility than Shleifer and Vishny do. Loewenstein and Willard (2006) also show that the model's violation of the law of one price is not robust to changes in its assumptions.

Gromb and Vayanos (2010) survey the theory and nest its main mechanisms in one model. A demand shock moves a price by an amount that rises with the arbitrageurs' risk aversion and with the part of the asset's risk that cannot be hedged. A cost of short selling adds a wedge only when arbitrageurs must be short, so the band around fair value becomes one-sided. Margin requirements make capital a constraint, and the gap between two identical assets closes only when arbitrageurs have enough of it. The model of Chapter 4 follows this structure and adds the feature that makes India unusual: a position limit that, once reached, lets every participant only reduce positions.

Two classic papers explain why short-sale constraints matter for prices. Miller (1977) argues that when opinions differ and pessimists cannot sell short, prices reflect the optimists and stocks are overpriced. Diamond and Verrecchia (1987) show that with rational traders constraints do not bias prices on average but slow their adjustment to private information, especially to bad news. Both predictions are tested in Chapter 8.

2.2. THE NO-ARBITRAGE BAND IN FUTURES MARKETS

A futures contract and the underlying stock bought with borrowed money give the same payoff, so the futures price should equal the cost of carry. Trading costs create a band around this value within which arbitrage does not pay. Brennan and Schwartz (1990) add that an arbitrage position carries an option to close it before expiry if the gap reverses; the model of this thesis uses the same idea.

Inside a band the basis should wander and outside it should revert, which calls for a threshold model. Dwyer, Locke and Yu (1996) estimate a three-regime threshold error-correction model of index futures and cash. Using minute data on the S&P 500 from 1982 to 1990, they find thresholds of about half a round-trip cost, faster reversion outside the band than inside, and no asymmetry between the two sides. Tse (2001) argues that arbitrageurs differ in their costs, so the aggregate response should rise smoothly with the deviation; his smooth-transition model of Dow Jones futures fits well, and reversion survives when observations outside the band are removed, so it is not an artefact of stale prices. Ters and Urban (2020) extend the threshold vector error-correction model of Hansen and Seo (2002) to three regimes with an unknown cointegrating vector and a constant. They show that the threshold and the constant are each poorly identified but their sum, which is the arbitrage cost, is estimated precisely, and they propose comparing the band before and after regulatory changes. Chapter 6 uses threshold and smooth-transition models in this tradition, on a daily panel of stocks rather than on one index pair.

2.3. SHORT-SALE CONSTRAINTS AND PRICES

Ofek, Richardson and Whitelaw (2004) price stocks twice, once directly and once through put-call parity. Violations point in the direction that short-sale costs predict: after shorting and trading costs, 13.6% of their observations lie above the upper bound and far fewer below it, violations are larger where borrowing is more expensive, and stocks that are expensive relative to their options later earn low returns. Stambaugh, Yu and Yuan (2015) use the asymmetry of arbitrage to explain why stocks with high idiosyncratic volatility earn low returns. With a mispricing score built from eleven anomalies, they show in double-sorted portfolios that idiosyncratic volatility goes with lower returns among the most overpriced stocks (-1.77% a month for high minus low) and with higher returns among the most underpriced (+0.52%), the first effect being larger. The anomaly test of Chapter 8 follows their design with anomalies that can be built from public Indian data.

Constraints can also be loosened. Gagnon (2018) studies about 1,300 introductions of single-stock futures in the United States and reports that they lowered the use and the fees of securities lending, a sign of looser short-sale constraints. For the speed of price adjustment I use the

TABLE 2.1

Earlier Indian estimates describe intraday bands; this thesis measures the overnight band

Studies of single-stock futures against their underlying stocks on the National Stock Exchange

STUDY	SAMPLE	DATA	BAND
Aggarwal (2015)	141 stocks, Mar to Aug 2013	One-second order book	About −15 to +3 bp, per stock and day
Shankar, Sankar and Kiran Kumar (2019)	102 stocks, 2007 to 2009	Five-minute prices	−39 to −10 bp, weekly
Slivka, Shah and Wu (2011)	Cost calculation	Fee schedules	About twice the width in mature markets
This thesis	360 stocks, 2015 to 2026	Daily volume-weighted prices	−34.5 to +23.9 bp around each stock's median

Source: the studies cited; Chapter 6 for this thesis.

price-delay measure of Hou and Moskowitz (2005), which compares how much of a stock's response to market returns comes with a lag.

2.4. EVIDENCE FROM INDIA

The Indian basis has been studied mostly at the index level or in short intraday samples. Nandan, Agrawal and Bhargava (2014) find that Nifty futures closed below their cost-of-carry value on 71% of days from 2003 to 2012, with the gap shrinking over the decade; they do not account for trading costs. Shankar, Sankar and Kiran Kumar (2019) estimate weekly three-regime threshold models for 102 stock futures with five-minute data from 2007 to 2009. Both of their thresholds are negative (−39 and −10 bp), the band is about 28 bp wide, it narrowed slightly when securities lending was introduced in 2008, and the lower edge moves about twice as much as the upper with volatility, which points to a constraint on the short side. Aggarwal (2015) estimates threshold models per stock and per day for 141 stock futures with one-second order book data from March to August 2013. The average band runs from about −15 to +3 bp, it is wider and more asymmetric for less liquid stocks, a common factor explains about 17% of the variation in daily band widths, and that factor moved with funding liquidity, for example when the Reserve Bank of India tightened liquidity in July 2013. Slivka, Shah and Wu (2011) compute the full cost of arbitrage between a stock and its future and find a band about twice as wide as in mature markets, with most arbitrage done within the day.

These bands are intraday objects for traders who close positions within minutes. The daily basis studied here measures the part of mispricing that lasts overnight, where an arbitrageur must carry a position and pay delivery-based taxes, and so is wider. Roy and Chakraborty (2023), using daily data for ten stocks from 2011 to 2017, find that stock and futures prices are cointegrated

and that the cash market does most of the correction of short-run deviations. Saurav, Srivastava and Agarwalla (2026) study the stock-futures basis around earnings announcements. According to their abstract, the basis turns negative before announcements and reverses afterwards, most strongly when disagreement is high and short-sale constraints are tight; Chapter 7 finds the same pre-announcement pattern outside bans.

Two Indian studies of the cross-section bear on the anomaly tests. Harshita, Singh and Yadav (2018) find drift after earnings announcements for Nifty 500 stocks from 2002 to 2017, measured with buy-and-hold returns from the second to the sixty-fourth day after the announcement; the drift is not related to idiosyncratic volatility or illiquidity. Sehgal, Vasishth and Deisting (2024) find that a stock's maximum daily return over the last five trading days of a month predicts low returns the next month, but that high idiosyncratic volatility goes with *higher* returns in India, the opposite of the United States. Results from the United States therefore cannot be carried over to India without checking.

2.5. INVESTORS AND REGULATION

The demand that pushes stocks towards their position limits comes largely from speculation. Kumar (2009) shows that individual investors in the United States prefer lottery-like stocks, which are cheap, volatile and positively skewed, and that these stocks underperform. Barber and Odean (2000) show that the households that trade most earn the least: the most active fifth earned 11.4% a year net of costs against 17.9% for the market. In India, SEBI's study of individual traders in equity derivatives (SEBI, 2024a) found that 93% of more than one crore individual traders lost money over the three years to March 2024. SEBI changed the rules for index derivatives in October 2024, citing the rise in retail participation and in speculative volumes (SEBI, 2024b), and changed the measurement of open interest and the position limits of single stocks in May 2025 (SEBI, 2025); the second change is studied in Chapters 6 and 7.

2.6. WHAT THIS THESIS ADDS

The theory of limited arbitrage is mature, the band econometrics is standard, and Indian studies show that wide and asymmetric bands exist. To my knowledge no study has estimated the effects of the F&O ban, the rule under which every participant may only reduce positions. The thesis models this rule, derives what it implies for the correction and the level of the basis, and tests the predictions with a plan fixed in advance. It also estimates the band over a long daily panel of stocks rather than an index or a short intraday sample, relates it to dated changes in costs, capital and position limits, and uses entry into the F&O segment and a constraint index for all listed stocks to study short-sale constraints across the whole cash market.

CHAPTER 3

Institutional setting

The Indian rules that shape each limit to arbitrage: who may sell short, which stocks have derivatives, how the position limit and the ban work and how they were enforced, what trading costs, and where arbitrage capital comes from. Figure 3.1 places the dated changes on one time line.

3.1. SHORT SELLING AND SECURITIES LENDING

Under SEBI's short-selling framework (restated in a circular of 5 January 2024, in line with the 2007 framework), all classes of investors may sell short, but naked short selling is prohibited: every short sale must be delivered at settlement. Institutional investors may not day trade. Securities traded in the F&O segment are eligible for short selling. An overnight short in the cash market therefore needs borrowed stock, which comes from the securities lending and borrowing (SLB) segment. SLB covers F&O stocks and, since July 2013, other liquid stocks that meet the exchanges' criteria, but trading in it is thin: on a typical day a few dozen stocks are lent. In practice the overnight short in an Indian stock is a short single-stock future (or an options position), and this exists only for F&O stocks. Stock derivatives have been physically settled since the October 2019 expiry.

3.2. THE F&O SEGMENT

A stock enters the F&O segment when it meets SEBI's eligibility criteria (market capitalisation, traded value, order size, MWPL and delivery value; the criteria were tightened in 2018 and again on 30 August 2024) and leaves it after failing them. From 2015 to October 2026 there were 239 entries and 171 exits, 162 of them eligibility exits after which the stock kept trading in the cash market. Entry gives a stock futures and options and formal short-selling eligibility at once, and exit removes them.

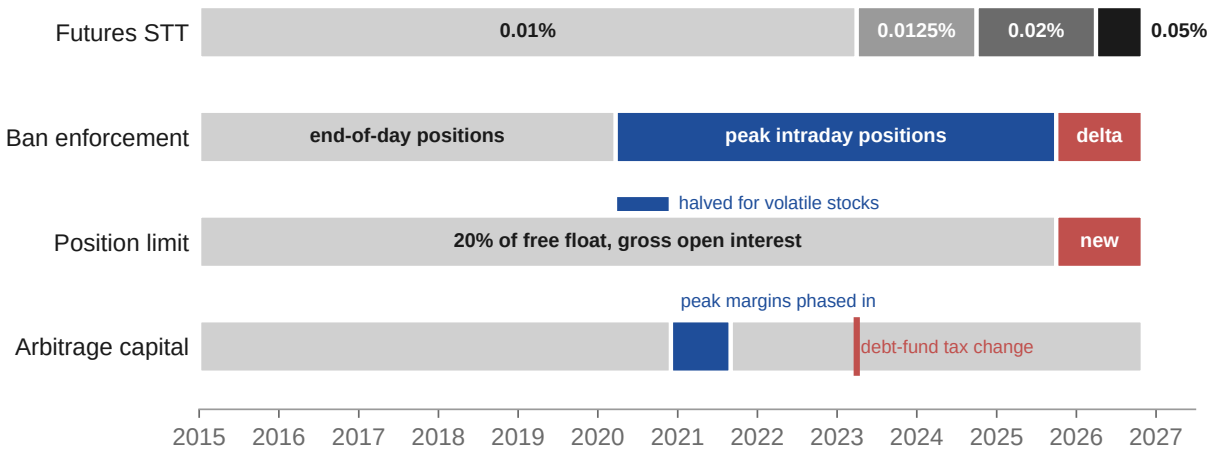
3.3. THE MARKET-WIDE POSITION LIMIT AND THE BAN

A stock enters the ban when market-wide open interest exceeds 95% of its MWPL at the end of a day, from the next day, and leaves it when open interest falls to 80%. During the ban every

FIGURE 3.1

Almost every limit to arbitrage changed on a known date during the sample

Futures STT, enforcement of the ban, the position-limit rule and arbitrage capital, 2015 to 2026



Source: SEBI circulars and press releases, NSE circulars (listed in the references).

TABLE 3.1

The limit was cut and open interest redefined in October 2025

Market-wide position limit (MWPL) and the open interest tested against it

PERIOD	MWPL	OPEN INTEREST TESTED
To 30 Sep 2025	20% of shares held by non-promoters (free float)	Gross open interest, futures and options
20 Mar to 26 Nov 2020	Halved for volatile or heavily used stocks (61 stocks)	Gross open interest
From 1 Oct 2025	Lower of 15% of free float and 65 times average daily delivery (in shares), floor 10% of free float; recomputed quarterly	Futures-equivalent (delta-adjusted) open interest

Source: SEBI circulars of 22 September 2006 and 29 May 2025; SEBI press release 18/2020; NSE Clearing circular 119/2025.

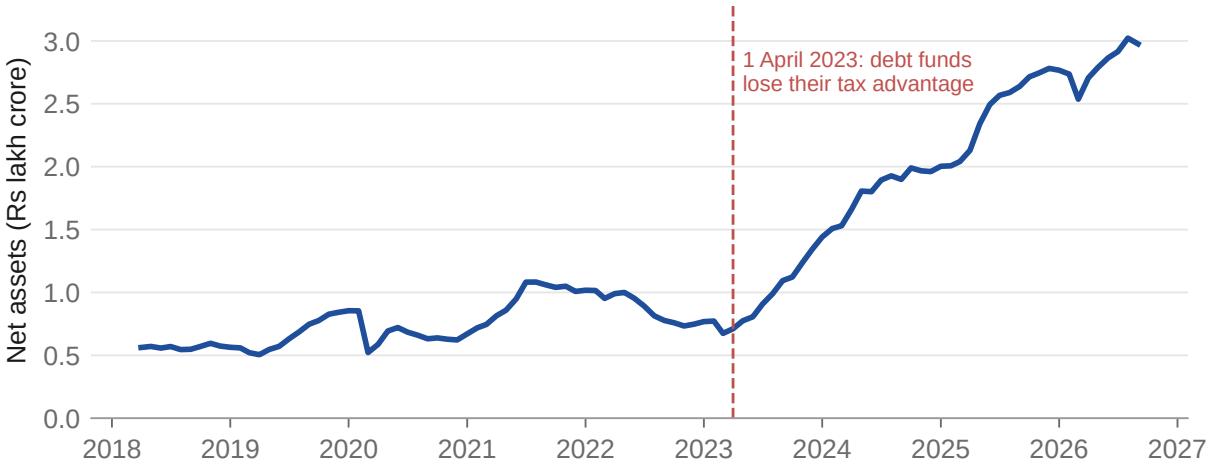
participant may only reduce positions, in all expiries and in both futures and options; rolling a position into the next expiry is not permitted. How the rule was enforced changed twice, and the changes matter for the results:

- *Until 22 March 2020*, violations were judged on end-of-day positions, so positions opened and closed within a day were not breaches.
- *From 23 March 2020*, after SEBI's press release 18/2020, the clearing corporations judged violations on peak intraday positions; this was kept "till further directions" when the other March 2020 measures were withdrawn in November 2020.
- *From 1 October 2025*, under SEBI's circular of 29 May 2025 and NSE Clearing circular 119/2025, an entity's end-of-day futures-equivalent position must fall relative to its position when the ban began. This is more permissive than the earlier rule of trading only through

FIGURE 3.2

Arbitrage-fund assets quadrupled after debt funds lost their tax advantage

Monthly net assets of the arbitrage-fund category, April 2018 to September 2026



Source: AMFI monthly reports; author's calculations.

offsetting positions: new contracts that lower the delta (for example a long put against a long future) are allowed. The package as a whole was not a relaxation: every stock's MWPL fell, and intraday monitoring of MWPL utilisation began on 3 November 2025.

Increases in a banned stock are penalised (from October 2025: 1% of the value of the increase, minimum Rs 5,000 and maximum Rs 1 lakh per entity per stock per day).

3.4. COSTS

The STT on the sale of futures was 0.01% from June 2013, 0.0125% from 1 April 2023, 0.02% from 1 October 2024 and 0.05% from 1 April 2026. Delivery trades in the cash market pay 0.1% on each side, and physical settlement of stock derivatives pays 0.1%. Stamp duty on purchases has been uniform since July 2020 (0.015% for delivery, 0.002% for futures). Exchange transaction charges were revised on 1 October 2024.

3.5. ARBITRAGE CAPITAL

Arbitrage mutual funds hold long cash, short futures positions and are taxed as equity funds. When the taxation of debt funds changed on 1 April 2023 (the same day as an STT increase), arbitrage funds became the tax-efficient home for short-term money. Their net assets were Rs 56,218 crore in April 2018, Rs 71,106 crore in April 2023, Rs 1.99 lakh crore in October 2024 and Rs 2.98 lakh crore in September 2026 (Figure 3.2).

3.6. PRICES

NSE sets a stock future's daily settlement price at a theoretical cost-of-carry value when the contract does not trade in the last half hour. A settlement-price basis would therefore shrink mechanically on thin days, including ban days; all measures in this thesis use traded prices. From 3 August 2026 the closing price of F&O stocks comes from a closing auction.

CHAPTER 4

A model of constrained arbitrage in India

One model in which every Indian limit to arbitrage is a parameter, written before the empirical work so that the hypotheses are derived rather than fitted. This chapter states the results; Appendix C gives the proofs and numerical checks.

4.1. SET-UP AND THE NO-ARBITRAGE BAND

The basis is the gap between the futures price and its cost-of-carry value, $m = \ln F - \ln F^*$ with $F^* = (S - \mathrm{PV}(D))e^{r(T-t)}$. Outside traders trade the stock and the future. Without arbitrage the basis equals u/Φ, where u is the relative demand pressure toward futures and Φ measures the depth of the two markets. An arbitrage sector holds a units of the cash-and-carry trade (long stock, short future). Market clearing gives

$$m = \frac{u - a}{\Phi}.$$

The sector chooses a to maximise $am - \frac{\gamma}{2}a^2 - c_+a^+ - c_-a^-$, where $\gamma = \alpha\sigma^2$ reflects risk aversion per unit of capital and the risk of the basis before convergence, c_+ is the cost of a cash-and-carry position and c_- the cost of a reverse position, which needs borrowed stock.

Result 1 (band). Arbitrage starts only when the basis leaves $[-c_-,\ c_+]$. Inside, the basis moves one-for-one with demand ($\partial m/\partial u = 1/\Phi$); outside, arbitrage absorbs part of each shock (slope $\gamma/(\gamma\Phi + 1)$). Figure 4.1 draws this.

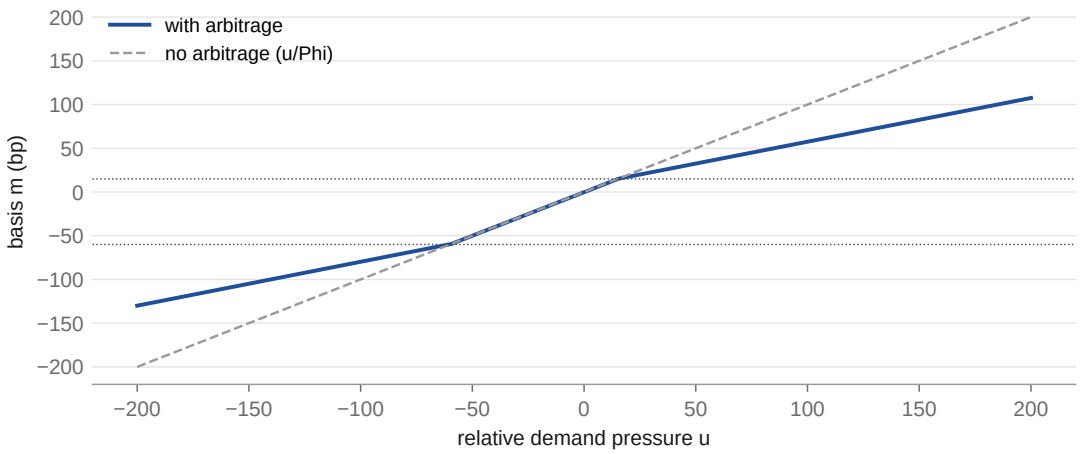
Result 2 (comparative statics). Costs set the edges one-for-one, so an STT increase should move the upper edge by about the change in cost; capital and risk set how far the basis travels beyond an edge; the edges shrink as expiry approaches; and because reverse arbitrage is costlier in India ($c_- > c_+$), the band is centred below zero. With cheap, tax-advantaged cash-and-carry capital whose required return is below the benchmark rate, even the upper edge can be negative.

Result 3 (inventory). When the sector already holds a_{t-1} and pays c_u to unwind, it trades only when the basis leaves $[\gamma a_{t-1} - c_u,\ \gamma a_{t-1} + c_+]$. Thresholds move with the size of the existing book. A threshold model with fixed thresholds therefore estimates average edges, not costs; I confirmed this on data simulated from the model.

FIGURE 4.1

Inside the band the basis follows demand one for one; outside it arbitrage absorbs part of each shock

Equilibrium basis against the demand pressure of outside traders, with illustrative costs $c_+ = 15$ bp and $c_- = 60$ bp



Source: model of Appendix C, Proposition C.1; illustrative parameters, not estimates.

4.2. THE F&O BAN AS A REDUCE-ONLY RULE

In a ban, (B1) arbitrageurs may only reduce, $0 \le a_t \le a_{t-1}$; and (B2) outside demand for *new* futures exposure cannot be executed: a share χ of it moves to the cash market (a trader who cannot buy the future buys the stock), and the rest is withdrawn, while closing demand still trades.

Result 4 (one-sided correction). A future that is too rich is no longer corrected, because nobody may open new cash-and-carry positions: every increase in pressure passes through fully. A future that is too cheap is still corrected by unwinding, at a floor $\gamma a_t - c_u$ that rises with the remaining book; the floor wears out as positions are unwound, and an expiry inside the ban removes positions that cannot be rolled. Figure 4.2 shows one simulated path.

Result 5 (sign reversal). In normal times new long demand raises the basis; in a ban the same demand lowers it, because part of it moves to the stock.

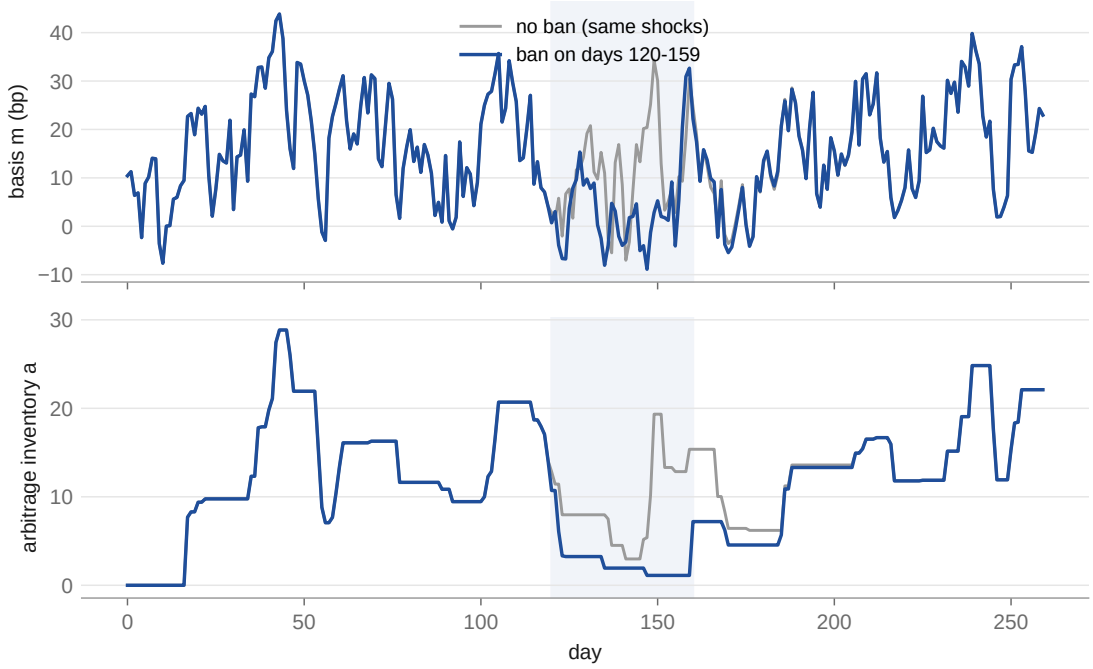
Result 6 (level effect). Relative to normal times, the ban raises the basis when pressure comes from closing trades and lowers it when new speculative demand keeps arriving. Its sign is not fixed; the observable condition is the composition of order flow.

Result 7 (trading). If the rule binds contract by contract, every futures trade in a ban closes a long and a short, so the change in open interest equals minus the volume; futures volume falls. Trades that offset across instruments (allowed from October 2025) break the identity.

FIGURE 4.2

In a ban arbitrage inventory can only fall, so upward pressure on the basis passes through in full

One simulated path with and without a ban on days 120 to 159, same shocks; basis (top) and arbitrage inventory (bottom)



Source: model of Appendix C; illustrative parameters, not estimates.

4.3. ANTICIPATION, HYSTERESIS AND IDENTIFICATION

Result 8 (anticipation). If exit is costlier in a ban, the effective cost of opening cash-and-carry is $c_+ + \pi(\omega)\ell$, where $\pi(\omega)$ is the chance that a position opened at utilisation ω meets a ban. The upper edge rises smoothly as utilisation approaches 95%; the jump at the trigger contains only the reduce-only effects. The model does not sign a density test: arbitrageurs holding back thin the density below 95%, speculators rushing in do the opposite.

Hysteresis. Between 80% and 95% a stock may be in a ban or not; at equal utilisation the difference is the effect of the reduce-only state against the anticipation state.

Result 9 (no instrument). The MWPL enters the ban state, the anticipation term and entity limits. A change in the MWPL moves the basis even without a ban, so it fails the exclusion restriction for an instrument of ban status; denominator changes identify the effect of the whole position-limit regime.

TABLE 4.1

Every hypothesis comes from the model with the result that would contradict it

The first four form the primary family; the rest are secondary

	PREDICTION	CONTRADICTED IF
H7	In a ban, rich futures revert more slowly; cheap futures as fast or faster (Result 4)	No asymmetry, or the opposite
H8	In a ban, futures volume falls and change in OI $\approx -$volume; cash activity rises (Result 7)	Futures activity unchanged
H3	Band edges move with the STT, more for larger increases (Result 2)	Edges unrelated to cost changes
H6	F&O entry lowers price delay and bad-news drift; exit raises them (Result 11)	No change or the opposite
Level	The level effect follows the composition of order flow (Result 6)	Sign unrelated to composition
H9	Upper tail rises smoothly toward 95%; reforms act through utilisation (Results 8, 9)	Jump only at 95%
H2	Drift after bad news larger in a ban (Result 10)	Symmetric response
H1	Wider bands where frictions are higher (Results 1 to 3)	No relation
H4	Scarcer capital widens the band (Result 2)	No relation
H5	Overpricing-side anomalies stronger where shorting is costly (Result 11)	Symmetric effects

Source: model of Appendix C; the frozen test plan of Appendix A.

4.4. SHORT-SALE CONSTRAINTS BEYOND THE BASIS

Result 10 (bad news in a ban). If a share ρ of informed selling runs through short futures, a ban lowers the share of bad news impounded on the day from k to $k(1-\rho)$, so drift after bad news rises; good news is unaffected.

Result 11 (Miller). With dispersed valuations and costly shorting, prices reflect optimists; the overpricing with no shorting is $(\sqrt{\delta} - \sqrt{Q/k})^2$, increasing in the dispersion δ. Entry into the F&O segment lowers the cost of shorting, so it should lower overpricing and speed the incorporation of bad news (Diamond and Verrecchia, 1987); exit should do the reverse.

4.5. FROM RESULTS TO HYPOTHESES

H7, H8, H3 and H6 form the primary family, corrected jointly for multiple testing; the rest are secondary and reported without adjustment.

CHAPTER 5

Data and measurement

All data come from public exchange archives and are built by scripts that check each step against the exchange's own rules. This chapter describes the sources and checks, the bans, the futures-cash basis, the other outcomes and a constraint index for every listed stock.

5.1. SOURCES

All data come from public archives, downloaded and parsed with scripts so that every step can be repeated (Appendix B).

5.2. CHECKS

Of the 11,764 daily "no fresh positions" flags, 11,753 (99.9%) match the next day's official ban list; utilisation exceeded 95% on the day before 99.4% of ban spells began and was at or below 80% on the last day of 99.7%. On three days the official list is missing at the exchange; those lists are imputed from the previous day's flags and continuing spells and flagged. The set of stocks in the open interest files matches the stock futures in the F&O bhavcopy, and an independent daily membership file for 2016 to 2025 agrees on every stock-day. The futures volume-weighted average price lies inside the day's range for 99.8% of contracts after recovering lot sizes in the older files. Results announcements are dated from the exchange's first dissemination time; the results filings alone often carry later re-filing times and current rather than historical symbols, so announcements, board meetings and the history of each security's identifier are combined, which covers 98% to 100% of F&O stock-quarters.

5.3. BANS

Over 2015 to 2026 there were 1,788 ban spells covering 9,416 stock-days and 141 stocks. Bans are concentrated: the most frequently banned stock had 76 spells. Before October 2025 the median spell lasted four trading days; the 44 spells since then have a median of nine. On 1 October 2025 the new formula lowered the MWPL of all 208 F&O stocks, to a median of 73% of the old value (Figure 5.2).

TABLE 5.1

Every series comes from a public archive

Sources, content and coverage

SOURCE	CONTENT	COVERAGE
NSE combined open interest files	MWPL, open interest, futures-equivalent open interest (from Oct 2025), next-day status	Feb 2010 to Oct 2026
NSE securities-in-ban lists	Official ban list	Feb 2010 to Oct 2026
NSE F&O and cash bhavcopies, delivery files	Prices, traded value and quantity, contracts, open interest, delivery	2015 to Oct 2026
NSE corporate actions	Dividends, splits, bonuses and other events with ex-dates	2014 to 2026
NSE announcements, results filings, board meetings	First dissemination time of quarterly results	2014 to 2026
NSE SLB bhavcopies	Daily securities-lending trades and fees	2015 to Oct 2026
AMFI	Half-yearly average market capitalisation lists; monthly arbitrage-fund flows and assets	2017 to 2026; Apr 2018 to Sep 2026
OECD via FRED	Three-month interbank rate (monthly)	2011 to 2026

Source: National Stock Exchange of India archives; Association of Mutual Funds in India; OECD Main Economic Indicators via FRED.

5.4. THE BASIS

For each stock and day, the basis is computed from the volume-weighted average prices of the future and of the stock, net of dividends going ex before expiry and of carry at the three-month interbank rate:

$$m_{it} = \ln \bar{F}_{it} - \ln \left(\bar{S}_{it} - \mathrm{PV}_{it}(D) \right) - r_{t\tau_{it}}.$$

The near-month contract is used, switching to the next month when three or fewer trading days remain; at least five contracts must trade; contracts with a dividend or other corporate event in their life and the first 20 trading days after a stock enters the F&O segment are excluded; the basis is winsorised at 1% and 99% within each year. The resulting sample has 486,040 stock-days for 371 stocks, 8,851 of them in a ban. Its median basis is +1.1 bp, and the 10th and 90th percentiles are −39.1 and +26.8 bp: the lower tail is wider than the upper (Figure 5.3), as expected when reverse arbitrage is costlier than cash-and-carry. The basis shrinks toward expiry

TABLE 5.2

Bans are frequent, concentrated in a few stocks, and longer under the new rule

F&O stocks, banned stocks, ban spells and ban stock-days by year

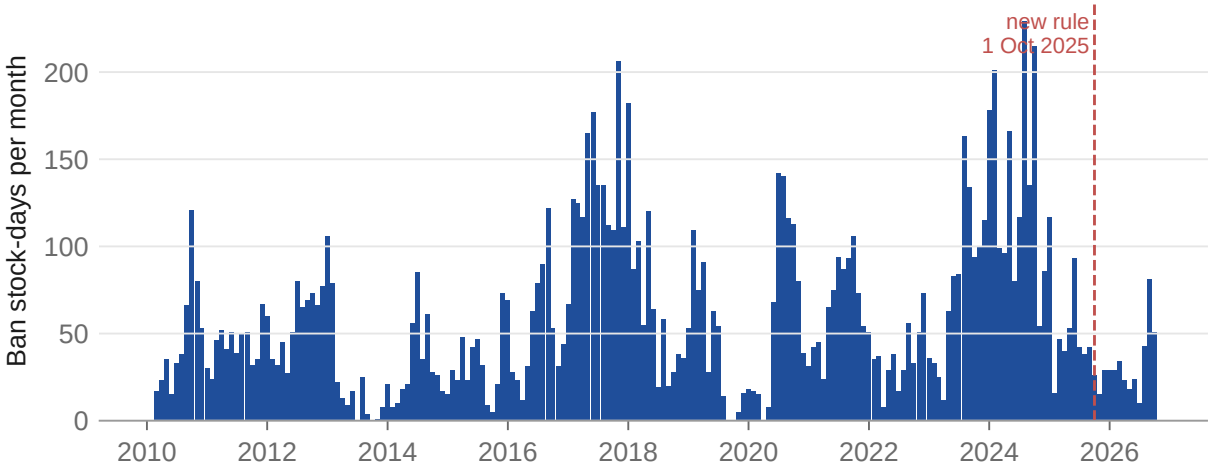
Year	F&O stocks (avg per day)	Stocks banned	Ban spells	Ban stock-days	Median spell (days)	Longest spell (days)
2015	154	16	69	367	4	21
2016	172	27	136	645	4	17
2017	200	43	287	1586	5	24
2018	206	38	162	810	4	31
2019	176	23	86	508	4	17
2020	140	40	178	756	3	13
2021	163	26	162	789	4	16
2022	196	23	100	457	4	13
2023	187	41	193	942	4	17
2024	185	49	289	1656	5	22
2025	216	37	93	558	5	16
2026[a]	209	10	33	342	10	29
2015 to 2026	183	141	1788	9416	4	31

[a] January to 27 October 2026. Spells are counted in the year they start. Source: NSE securities-in-ban lists and F&O bhavcopies; author's calculations.

FIGURE 5.1

Bans come in waves and became rarer under the new rule

Number of stock-days in the ban per month, February 2010 to October 2026



Source: NSE securities-in-ban lists; author's calculations.

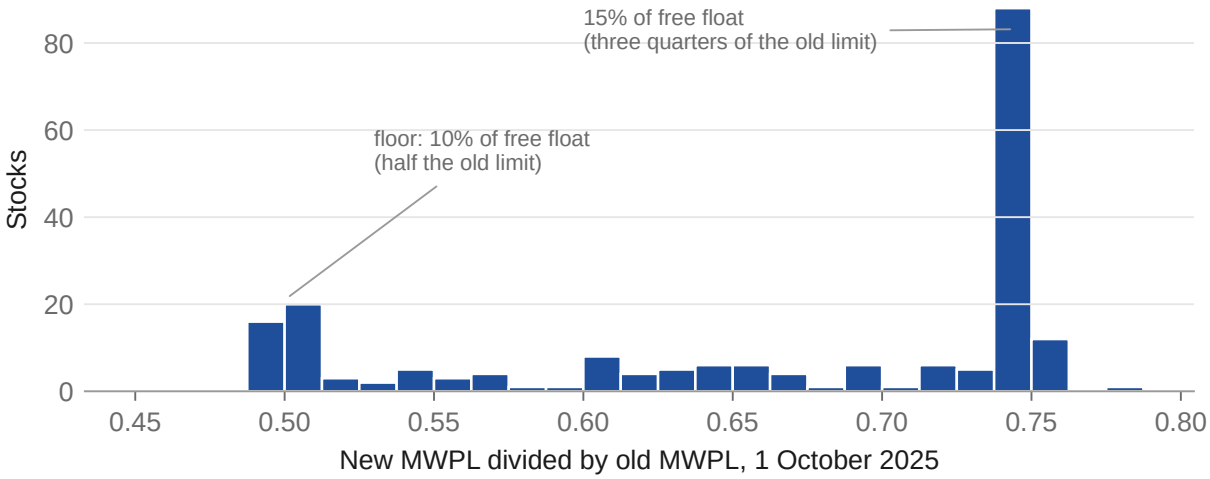
(Figure 5.4).

A relative implied repo rate (the stock's implied financing rate minus the day's cross-sectional median) and a basis built from closing prices are used as checks.

FIGURE 5.2

The October 2025 formula cut every stock's limit, by a quarter to a half

Ratio of the new to the old MWPL on 1 October 2025, all 208 F&O stocks

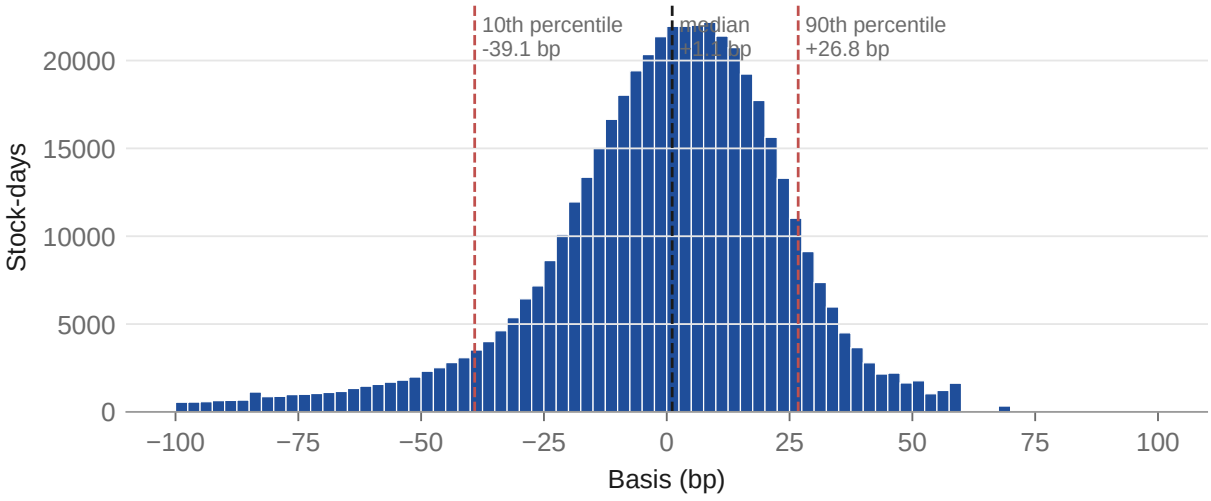


Source: NSE combined open interest files of 30 September and 1 October 2025; author's calculations.

FIGURE 5.3

The basis is centred near zero with a long lower tail

Basis of the analysis sample, 2015 to 2026, in basis points (3.1% of stock-days lie beyond 100 bp from zero and are not drawn)



Source: NSE F&O and cash bhavcopies, corporate actions, three-month interbank rate; author's calculations.

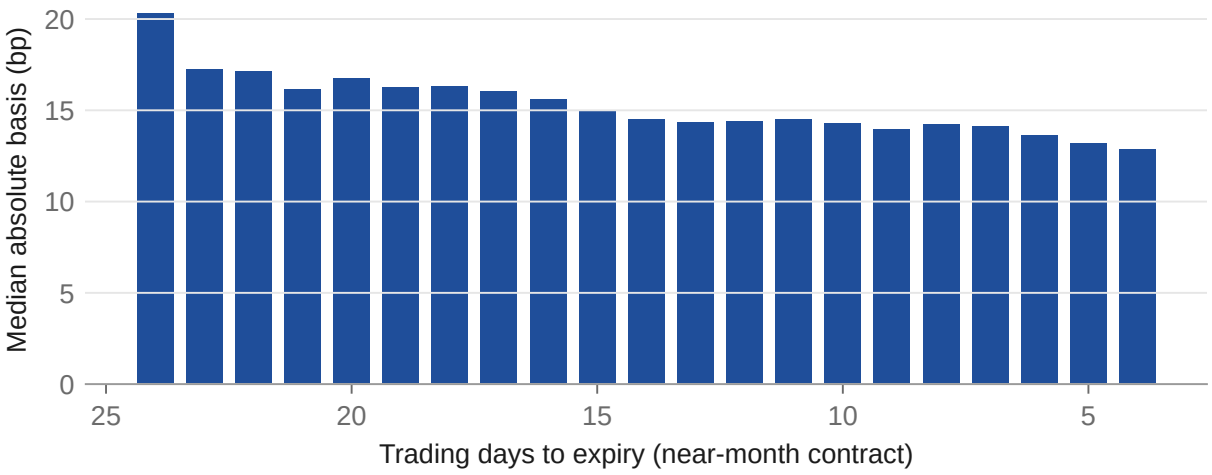
5.5. OTHER OUTCOMES

- *Trading:* futures contracts traded (near and next month), the change in open interest in contracts, the ratio $R = (\Delta OI + V)/V$, cash volume and the delivery share.
- *Earnings drift:* the buy-and-hold return over days +2 to +64 after a results announcement minus that of the stock's size-decile portfolio; the surprise is the abnormal return over days −1 to +1; bad (good) news is the bottom (top) tercile within each quarter. Returns are adjusted for splits, bonuses and dividends (the exchange's previous-close field is not), and

FIGURE 5.4

The basis narrows as expiry approaches

Median absolute basis of the near-month contract by trading days to expiry (in the last three days the sample uses the next contract)



Source: as for Figure 5.3.

events with a price below Rs 5 are excluded.

- *Price delay:* the share of a stock's weekly return variation explained by lagged market returns (Hou and Moskowitz, 2005), by calendar year.

5.6. A CONSTRAINT INDEX FOR ALL STOCKS

To measure how hard arbitrage is in each stock without using the outcomes or the treatments, I build a quarterly index for all NSE stocks (80,042 stock-quarters, 3,310 stocks, 2015 to 2026) from five frictions: no short channel (not in the F&O segment), SLB scarcity (share of days without any SLB trade), the Corwin and Schultz (2012) spread, Amihud (2002) illiquidity and idiosyncratic volatility. Each is standardised within the quarter, and the index is their first principal component, which explains 53% of their variance; all five load positively (0.37 to 0.49) and the loadings are stable from year to year. The index is persistent (quarter-to-quarter rank correlation 0.90).

More constrained stocks respond more slowly to market news, also within a stock over time and after controlling for size, and drift more after announcements in either direction (Table 5.3). Without a size control, drift after bad news, but not good news, is stronger for more constrained stocks; with size-decile fixed effects this asymmetry disappears, so it reflects the smaller size of constrained stocks rather than the short-sale constraint. These are associations. Within F&O stocks, the short-channel component does not vary and SLB activity appears to reflect demand to borrow rather than ease of borrowing, so for the futures-cash band of F&O stocks I use a separate F&O index that adds utilisation of the position limit (Chapter 6).

TABLE 5.3

More constrained stocks adjust more slowly, also within stocks and after controlling for size

Coefficients per standard deviation of the index, frictions measured before the outcomes

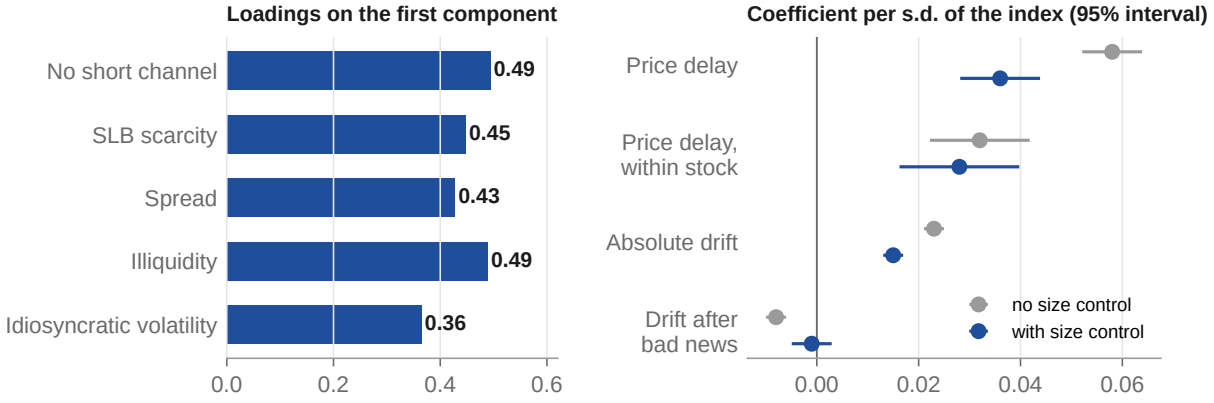
OUTCOME	PER S.D. (S.E.)	SIZE-CONTROLLED	OBS.
Price delay, next year (year FE)	+0.058 (0.003)	+0.036 (0.004)	15,790
Price delay, next year (stock and year FE)	+0.032 (0.005)	+0.028 (0.006)	15,790
Absolute drift after announcements	+0.023 (0.001)	+0.015 (0.001)	69,673
Drift after bad news	−0.008 (0.001)	−0.001 (0.002)	23,219
Drift after good news	+0.001 (0.001)	+0.001 (0.002)	23,095

FE: fixed effects. Size control: log market capitalisation for price delay (15,227 stock-years with a market capitalisation); market-capitalisation decile fixed effects for drift. Source: author's calculations.

FIGURE 5.5

All five frictions load on one component, and its link with price delay survives a size control

Loadings on the first principal component (left) and validation coefficients with and without a size control (right)



Source: NSE bhavcopies and SLB files, AMFI market capitalisation; author's calculations.

CHAPTER 6

The no-arbitrage band: costs, capital and position limits

The band within which the basis moves in normal times, and how it relates to the three slow-moving limits: trading costs, arbitrage capital and the position-limit regime. The chapter uses non-ban stock-days only, except where it counts bans.

6.1. ESTIMATING THE BAND

Let x_{t-1} be yesterday's basis minus the stock's median basis on non-ban days, in basis points, and Δm_t the change in the basis of the same contract. The three-regime threshold model is

$$\Delta m_t = a_r + \rho_r x_{t-1} + e_t, \qquad r = \begin{cases} \text{lower} & x_{t-1} < g_L \\ \text{middle} & g_L \le x_{t-1} \le g_U \\ \text{upper} & x_{t-1} > g_U, \end{cases}$$

with the thresholds chosen to minimise the sum of squared residuals over a grid of percentiles of x (at least 10% of observations in each outer regime and 20% in the middle). Linearity is tested with a sup-F statistic whose distribution under the linear model comes from a wild bootstrap clustered by date; threshold intervals come from resampling whole stocks. I also estimate exponential and logistic smooth-transition models (Tse, 2001), in which the speed of reversion changes smoothly with the deviation, and models in which it changes smoothly with an observed state.

Linearity is rejected (sup-F 7,471 against a bootstrap 95% critical value of 31). The band is about 58 bp wide and asymmetric: the lower edge is further from the centre, as expected when reverse arbitrage is costlier than cash-and-carry. By the Bayesian information criterion the threshold model fits best, the logistic smooth-transition model (slow reversion below +5 bp, fast above) almost as well, and the symmetric exponential model clearly worse; a symmetric smooth transition does not suit an asymmetric band.

The slopes need care. Inside the band the model of Chapter 4 predicts almost no correction, yet the middle slope implies a half-life of about one day. Daily volume-weighted prices contain noise, and classical noise biases these slopes toward -1. Instrumenting x_{t-1} with x_{t-2} (last row of Table 6.1, with regimes classified by x_{t-2}) shrinks all three slopes and removes most of the difference between the upper and lower regimes. Much of the day-to-day reversion is therefore noise, and the slower reversion of cheap futures in the threshold model is not a robust finding. The thresholds, which describe where the basis spends its time, are less affected.

TABLE 6.1

The basis moves in a wide, asymmetric band, but much of the daily reversion is noise

Pooled three-regime threshold model, non-ban days, 2015 to 2026 (473,110 observations, 360 stocks)

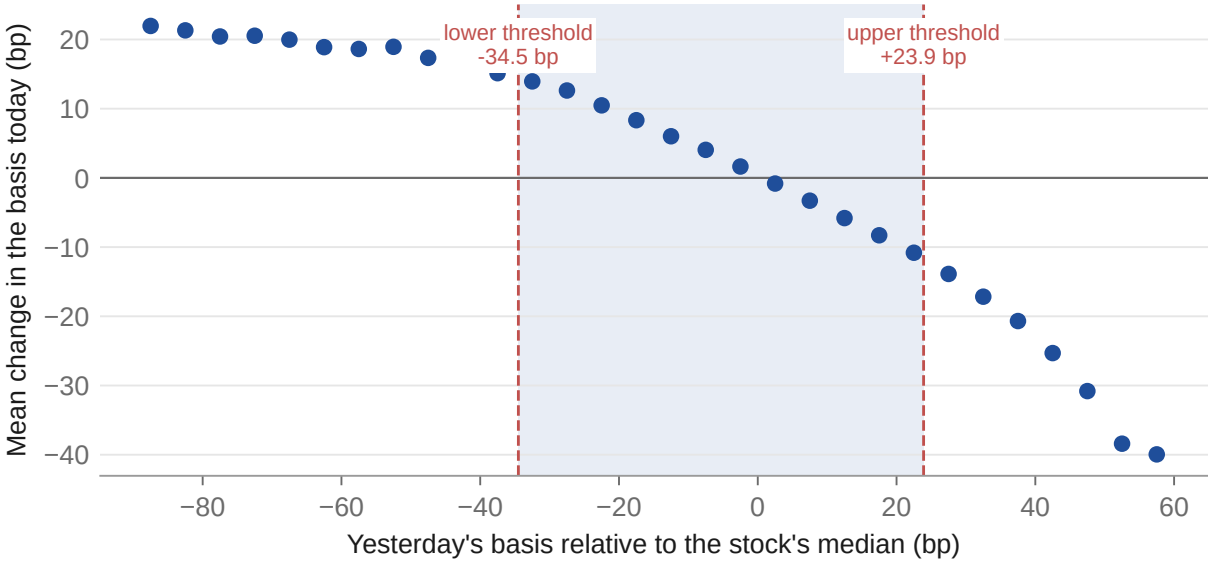
	LOWER REGIME	MIDDLE REGIME	UPPER REGIME
Threshold (bp, relative to the stock's median)	below −34.5		above +23.9
95% interval (stock bootstrap)	[−37.2, −32.2]		[+20.3, +26.1]
Share of observations	0.12	0.76	0.12
Slope of Δm_t on x_{t-1}	−0.084	−0.468	−0.809
Slope, instrumented by x_{t-2}	−0.121	−0.192	−0.141

Thresholds chosen by least squares over a grid of percentiles; intervals from resampling whole stocks. Source: author's calculations.

FIGURE 6.1

The further the basis moves from the stock's median, the more it reverts the next day

Mean change in the basis today by yesterday's deviation from the stock's median (5 bp bins), non-ban days; shaded: the estimated band



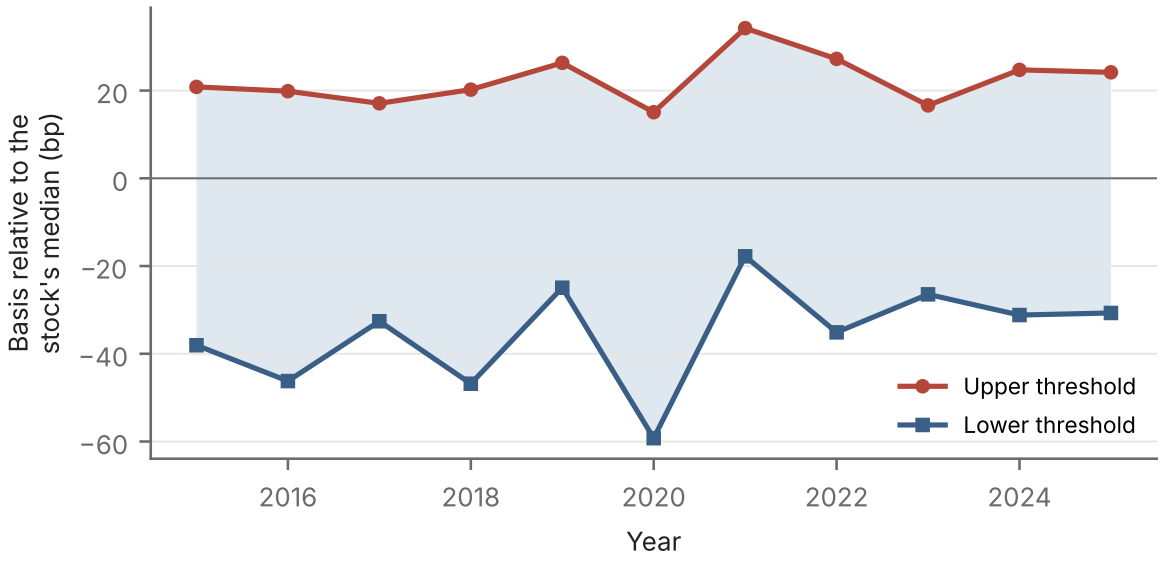
Source: author's calculations.

The band is stable across years (Figure 6.2), apart from 2020, when the March dislocation pushed the lower edge to −59 bp. The lower edge is wider far from expiry (−40 bp with 17 or more trading days left against −28 bp in the last ten days), in line with edges that shrink as expiry approaches. In a smooth-transition model with the number of trading days to expiry as the state, reversion is faster in the last week (the slope rises by 0.22 above seven days, s.e. 0.02).

FIGURE 6.2

The band is stable from year to year, apart from the dislocation of March 2020

Thresholds of the pooled threshold model by calendar year, non-ban days, basis relative to each stock's median



Source: author's calculations.

TABLE 6.2

The upper part of the basis rose after the 2024 and 2026 tax increases

Mean change in each stock's percentiles of the basis, 40 trading days after against 40 before (bp)

EVENT	STT CHANGE	STOCKS	80TH PERCENTILE	20TH	MEDIAN
3 Apr 2023[a]	+0.25	181	−2.12	+0.24	+0.27
1 Oct 2024	+0.75	172	+6.42	+4.55	+5.39
1 Apr 2026	+3.0	192	+7.44	+2.10	+6.30

[a] Same day as the change in the taxation of debt funds; reported, not used as evidence. STT change in bp of the futures sale.
Source: author's calculations.

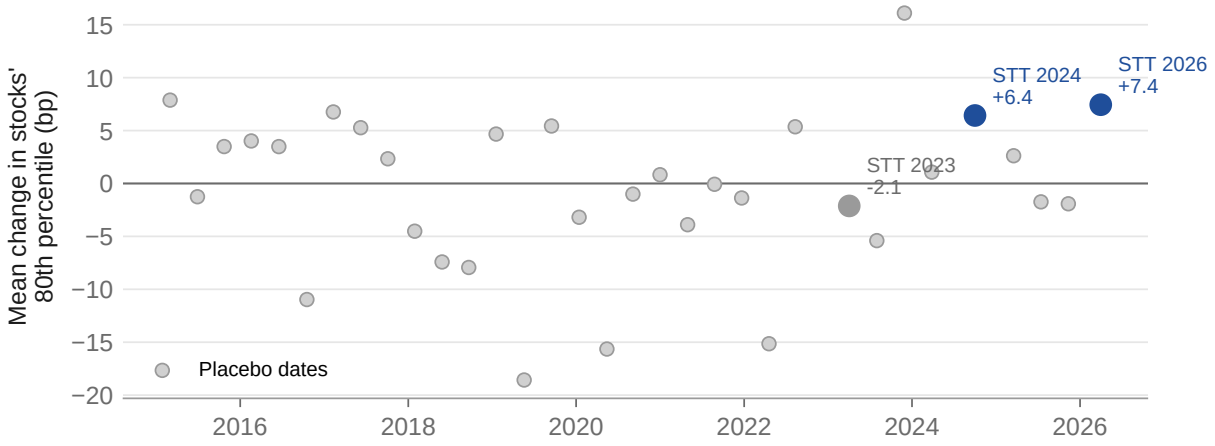
6.2. COSTS

Pre-specified test (H3). Around each STT increase I compare each stock's 80th percentile of the basis in the 40 trading days before and the 40 from the event. Pooling 2024 and 2026, the mean change in the 80th percentile is +6.96 bp (s.e. 0.61, clustered by stock), and the ordering of the three events matches the size of the cost changes. On the pre-specified statistic H3 is supported. The pre-specified placebo, the same statistic at 30 dates away from the true events, tells a more cautious story: shocks common to all stocks move the whole distribution, the placebo statistic has a standard deviation of 7.4 bp, and the true changes rank at the 10th and 7th percentiles of the placebo distribution (Figure 6.3). The changes in the band's width and in

FIGURE 6.3

Against placebo dates, the rise after the tax increases is large but not exceptional

Mean change in stocks' 80th percentile of the basis at the three STT dates and at 30 placebo dates (bp)



Placebo dates every 80 trading days, away from the true events. Source: author's calculations.

the upper percentile relative to the median are not unusual. The rise is mostly a shift of the whole distribution, larger than the cost changes, and standard errors clustered only by stock overstate the precision when every observation shares the same event date (Kolari and Pynnönen, 2010).

As a further check, the same statistic at the three dates of SEBI's 2024 to 2025 reforms of index derivatives, which did not change the costs of single-stock arbitrage, gives −4.1, −0.3 and +1.9 bp, all well inside the placebo range.

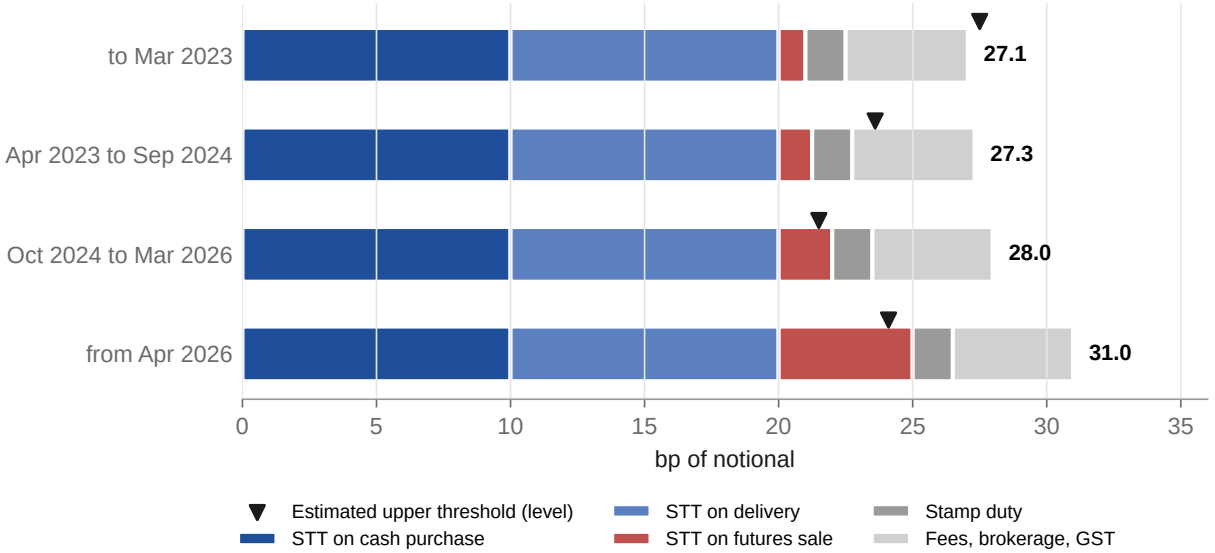
Band edges by tax period. Measuring each deviation from the stock's median within each period (the full-sample median does not remove the 2026 level shift), the upper threshold is +23.8, +18.9, +23.1 and +28.9 bp in the four STT periods and the lower threshold −39.1, −30.9, −31.5 and −31.5 bp. The upper edge is highest in the highest-tax period but does not rise step by step with the tax. Reversion above the upper edge weakens over time (slope −0.77, −0.55, −0.20, −0.15), which coincides with both rising costs and rising arbitrage capital.

A bottom-up cost band. The full round trip of an overnight cash-and-carry position (STT of 0.1% on the cash purchase and on delivery at expiry, STT on the futures sale, stamp duty, exchange charges, the SEBI fee, GST and an institutional brokerage of 0.01% per leg) costs 27.1, 27.3, 28.0 and 31.0 bp in the four periods (25 to 38 bp for brokerage between 0.005% and 0.03% per leg). Rolling an existing position for one more month costs 4.0 to 8.0 bp. The estimated upper threshold in levels (+22 to +28 bp) and the 80th percentile lie below the full round trip, which sits between the 90th and 95th percentiles. New capital enters only at large deviations; funds that keep capital in place and roll can lean against much smaller ones. This is one reason why fixed thresholds sit inside the full cost band, as the model's inventory result implies (Figure 6.4).

FIGURE 6.4

The full cost of a new position lies above the estimated upper edge

Round trip of an overnight cash-and-carry position held to delivery, by STT period (bp of notional); triangles: estimated upper threshold in levels



Institutional brokerage of 0.01% per leg; GST of 18% on fees and brokerage. Source: NSE circulars on STT and transaction charges; author's calculations.

6.3. ARBITRAGE CAPITAL

Pre-specified test (H4). Monthly, from May 2018 to September 2026, I regress the 80th percentile of the basis across all non-ban stock-days on last month's net inflow into arbitrage funds as a share of their assets, with step dummies for April 2023, October 2024 and April 2026. The coefficient is −17.5 bp per unit of flow (Newey-West s.e. 14.2, one-sided $p = 0.11$): a 10% inflow goes with a 1.7 bp lower upper quantile the next month. In first differences the relation is clearer (−20.6, s.e. 6.8), and it is visible before April 2023 (−26.0, s.e. 16.5) but not after (+17.1, s.e. 16.9). Same-month flows have a positive coefficient, as expected when flows chase the returns that a rich basis offers. With about a hundred months the test has little power (Figure 6.5).

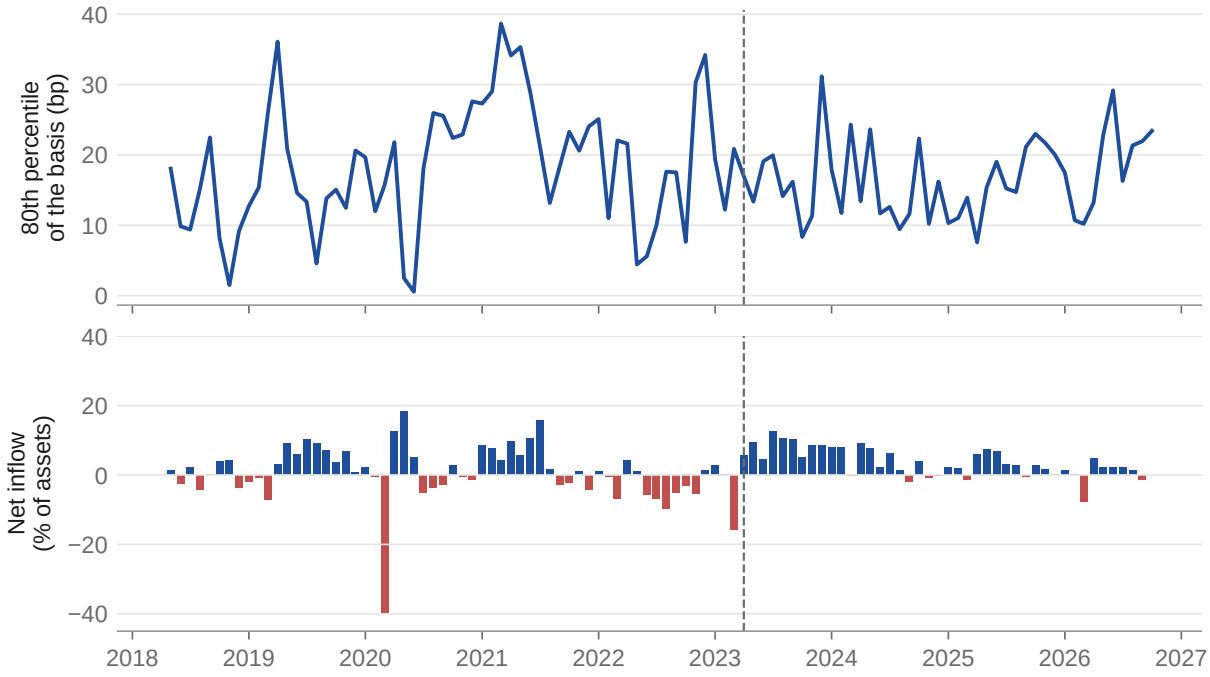
Capital as a state. In a smooth-transition model with log arbitrage-fund assets as the state, reversion is faster when assets are large (the slope falls by 0.077, s.e. 0.019, above about Rs 95,000 crore). This variable also orders time, so the evidence is descriptive.

Peak margins. From December 2020 to September 2021 margins had to be collected on peak intraday positions, in four steps of 25%. This raises the capital an arbitrageur must post during the day. The upper percentile moves in both directions at the four steps (+3.3, +9.6, −9.9 and +3.5 bp, against a placebo standard deviation of 7.4), so I find no consistent effect; the rule mainly binds intraday leverage, which a daily basis does not measure.

FIGURE 6.5

The upper tail of the basis and arbitrage-fund flows move a great deal from month to month

Monthly 80th percentile of the basis across non-ban stock-days (top) and net inflow into arbitrage funds as a share of assets (bottom); dashed: April 2023



Source: AMFI monthly reports; NSE bhavcopies; author's calculations.

6.4. POSITION LIMITS

Anticipation (H9). On non-ban days before October 2025, the share of days on which a stock's basis is in its own upper tail (above the 80th percentile of its days with utilisation below 60%) rises from 0.20 to about 0.27 as utilisation the day before approaches 95% (Figure 6.6). With stock and date fixed effects, the slope of this share on utilisation between 0.60 and 0.95 is +0.079 (s.e. 0.038, one-sided $p = 0.019$), the lower tail does not change, and the increase comes more as a step around 0.75 than as a smooth slope. The basis is more often rich as the trigger nears, as the anticipation result predicts, although high utilisation also signals demand pressure. In a smooth-transition model with utilisation as the state, reversion weakens as utilisation rises (the slope rises by 0.225, s.e. 0.053).

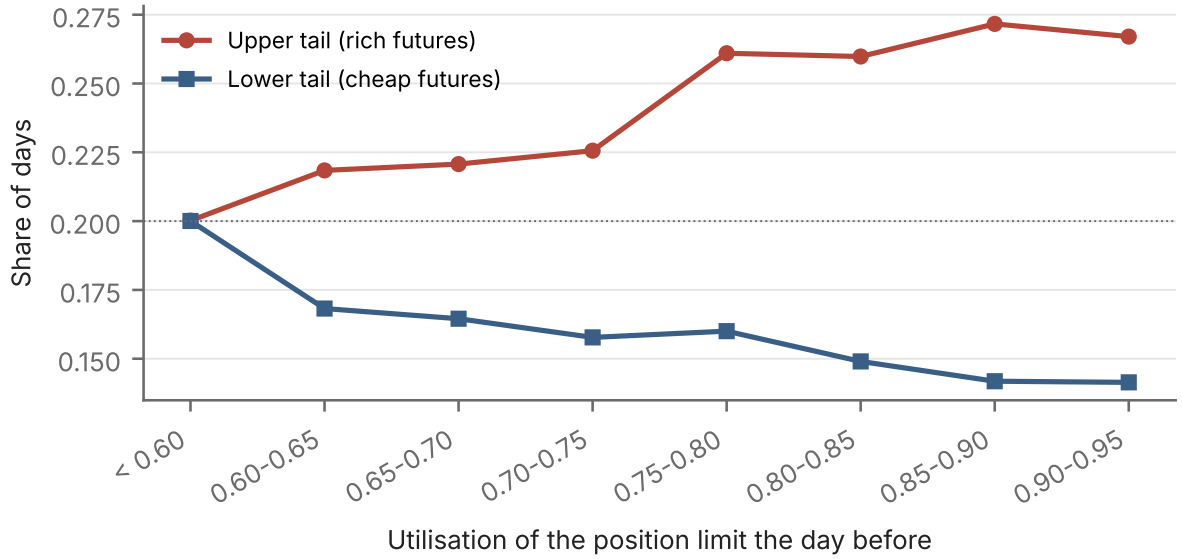
The October 2025 formula and quarterly resets. The size of each stock's MWPL cut on 1 October 2025 (0.24 to 0.69 log points) raised its utilisation strongly (first stage +0.28 per unit, s.e. 0.03). Its effect on the upper tail is +0.061 (s.e. 0.050, $p = 0.11$), on the level +4.8 bp and on bans +0.023: all in the predicted direction, none significant. The quarterly resets of January, April and July 2026 move utilisation but show no effect on the basis.

How the October 2025 rules changed the number of bans. SEBI's stated aim was to reduce "spurious" bans. Before the reform, bans were driven by open interest, not by changes in the limit: on the day utilisation crossed 95% the limit fell for only 8 of 1,744 entries, and only 7 (0.4%)

FIGURE 6.6

Rich futures become more common as utilisation approaches the 95% trigger

Share of non-ban days with the basis in the stock's upper and lower tail, by utilisation the day before (raw shares, before October 2025)



With date fixed effects the lower-tail decline disappears; the upper-tail rise remains. Source: author's calculations.

TABLE 6.3

The new rule produced about a quarter of the ban days the old rule would have

Bans from 1 October 2025 to 27 October 2026 for the stocks in the segment on 30 September and 1 October 2025

OPEN INTEREST AND LIMIT	BAN STOCK-DAYS	SPELLS	STOCKS
Delta-adjusted, new limit (the actual rule)	412	45	11
Gross, old-formula limit (the old rule)	1,505	112	36
Gross, new limit	6,256	296	85
Delta-adjusted, old-formula limit	1	1	1

The 45 spells include one that began on 30 September 2025. Simulated ban rule validated against the official lists. Source: NSE combined open interest files; author's calculations.

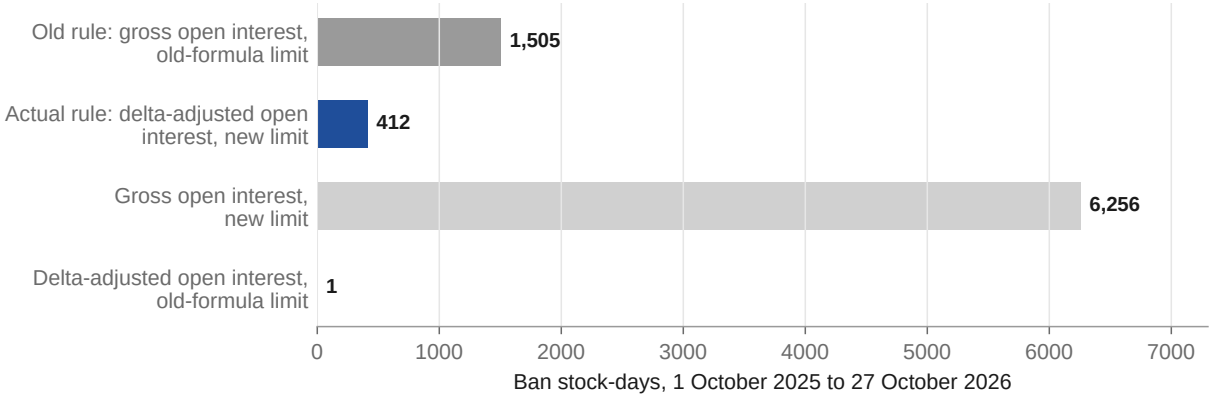
would not have crossed at the previous day's limit. To see what the two parts of the reform did, I simulate the ban rule (enter the day after utilisation exceeds 95%, leave the day after it is at or below 80%) for the stocks that were in the segment on both 30 September and 1 October 2025. Applied to the actual utilisation, the simulation reproduces the official ban lists exactly. I then apply it with open interest measured gross or in delta terms, and with the new limit or the limit the old formula would have given (each stock's new limit scaled by its ratio of old to new limit on 1 October 2025, an approximation).

Under the old rule the same stocks would have spent about 3.7 times as many days in a ban (Table 6.3). The two parts of the reform pull in opposite directions: measuring open interest in delta terms removes almost every ban, because options now count only by their delta and offsetting positions of the same client net out, while the lower limit on its own would have

FIGURE 6.7

Measuring open interest in delta terms removes almost all bans; the lower limit alone would have multiplied them

Ban stock-days, 1 October 2025 to 27 October 2026, under the actual and three counterfactual rules



Source: as for Table 6.3.

quadrupled them. The combination cut bans by about three quarters.

The 2020 halving and restoration. From 23 March 2020 the limit was halved for stocks whose recent daily price range or utilisation was high, and on 27 November 2020 it was restored for the 70 stocks whose limit was then halved. The restoration is a clean shock: it was announced, it came in a calm market, and it doubled the limit of some stocks and not others. Against other F&O stocks, in the 40 trading days either side, the utilisation of the restored stocks fell by 0.21 (s.e. 0.02), their share of days in a ban by 4.6 percentage points (s.e. 1.4), and the share of days with their basis in its upper tail by 6.8 percentage points (s.e. 3.2). The basis level did not change significantly (−2.8 bp, s.e. 3.1). A false restoration date 20 days earlier shows no effect on utilisation, bans or the upper tail. A looser limit makes rich futures less common, as the anticipation result predicts, although the upper-tail effect is weaker with a 20-day window (−6.1, s.e. 4.0). The halving itself coincided with the March 2020 crash and selected stocks on the outcomes, so I do not interpret it.

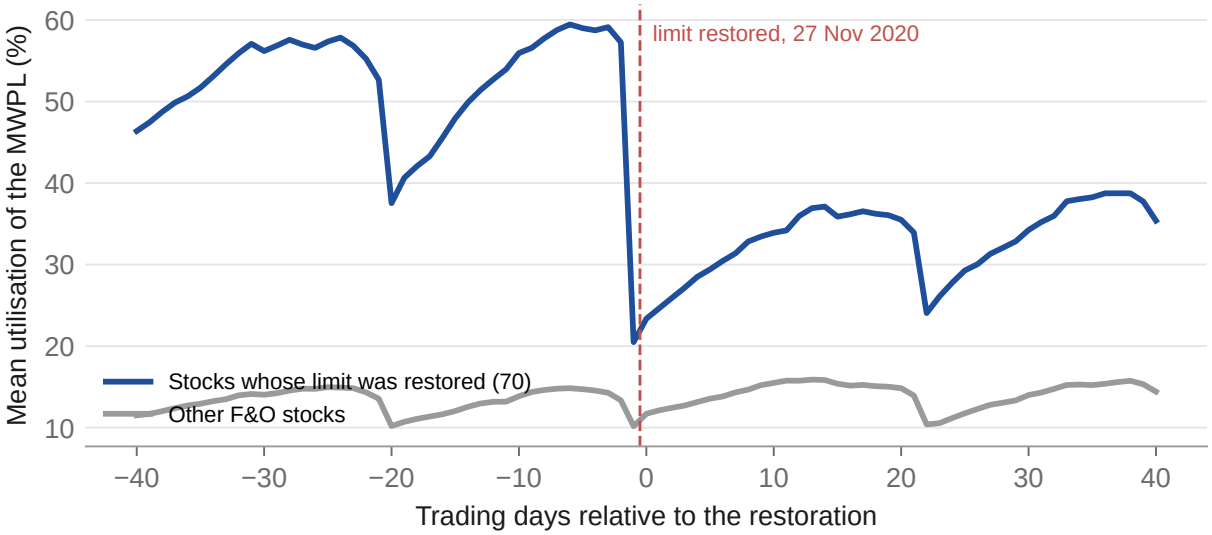
6.5. FRICTIONS ACROSS STOCKS (H1)

For F&O stocks I build a quarterly friction index from the spread, illiquidity, idiosyncratic volatility, utilisation of the position limit and SLB scarcity, measured in the previous quarter. A one standard deviation higher index goes with a band (80th minus 20th percentile of the basis) 3.5 bp wider (s.e. 1.1, one-sided $p = 0.002$; mean 40.8 bp) and a mean absolute basis 2.5 bp larger. The relation is across stocks: with stock fixed effects it disappears. The persistence measure moves the wrong way, which the noise result above explains, and SLB scarcity enters with the wrong sign, which suggests that SLB activity reflects demand to borrow. Per-stock threshold models (258 stocks) give a median band of −26 to +17 bp, and their widths are not related to the stocks'

FIGURE 6.8

Restoring the limit cut the utilisation of the affected stocks at once

Mean utilisation of the MWPL for the 70 stocks whose limit was restored on 27 November 2020 and for other F&O stocks



The fall on day 0 is mechanical (the denominator doubled); the tests use bans and the basis. Source: NSE combined open interest files; author's calculations.

average index.

6.6. SUMMARY

The basis follows a wide, asymmetric, nonlinear band. Its edges respond to costs less cleanly than the model implies; the full cost of new arbitrage lies in the outer tail, while the cost of keeping capital in place is small. Capital and the position-limit regime leave traces in the predicted direction (faster correction with more capital, a richer basis near the trigger, a less often rich basis after the 2020 restoration of the limit), but the formal tests of capital flows and of the 2025 reform are not significant. The 2025 rules cut the number of bans by about three quarters.

CHAPTER 7

The F&O ban

What happens to trading and to the futures-cash basis when a stock enters the ban: the pre-specified tests of trading and of one-sided correction, the level of the basis at the start of a ban, the 95% trigger, news in a ban, and placebo checks at false ban dates.

7.1. TRADING IN A BAN (H8)

I compare each stock's ban days with its own non-ban days at similar utilisation the day before (80% to 95%, the only range in which both occur), before October 2025, excluding expiry days, with stock-by-utilisation-bin and date fixed effects and standard errors clustered by stock and date. Futures trading falls by about 87%, cash volume rises by about 7.5% and more of it is delivered (Table 7.1). H8 is supported.

The model also predicts that if the rule binds trade by trade, the change in open interest equals minus the volume, so $R = (\Delta OI + V)/V$ is zero on ban days (it is about one on normal days). Over the whole period the ban-day mean of R is 0.42, so the identity fails on the pre-set reading. The explanation lies in enforcement (Figure 7.2): until 22 March 2020, when violations were judged on end-of-day positions, the median ban-day R is 0.87; from 23 March 2020, when the clearing corporations judged violations on peak intraday positions, it is 0.011; under the delta-based rule from October 2025 it is 0.19 to 0.36. When the rule binds trade by trade, the identity holds almost exactly. The effect of a ban on futures volume is -0.97 log points before March 2020 and -2.72 after (Figure 7.1).

7.2. DOES CORRECTION BECOME ONE-SIDED? (H7)

I regress the daily change in the basis on yesterday's deviation from the stock's median, separately in an upper regime (above the stock's 80th percentile on non-ban days), a middle and a lower regime (below the 20th percentile), and let each slope change on ban days. The model predicts that upward deviations revert more slowly in a ban ($\delta_{up} > 0$) and downward deviations as fast or faster. The prediction fails, with the opposite sign: in a ban rich futures revert faster and cheap futures more slowly (Table 7.2). This holds with other threshold definitions, without imputed ban days and with the relative implied repo rate; with the close-based basis the upper change is about zero. It is stronger on liquid days, differs by enforcement period, and is concentrated in the first days of a spell (-0.163 in the first five calendar days against -0.008 later). Instrumenting yesterday's deviation with the deviation two days earlier, to remove the effect of noise in daily

TABLE 7.1

In a ban, futures trading collapses and cash trading moves toward delivery

Ban days against the same stock's non-ban days at utilisation of 80% to 95% the day before, before October 2025

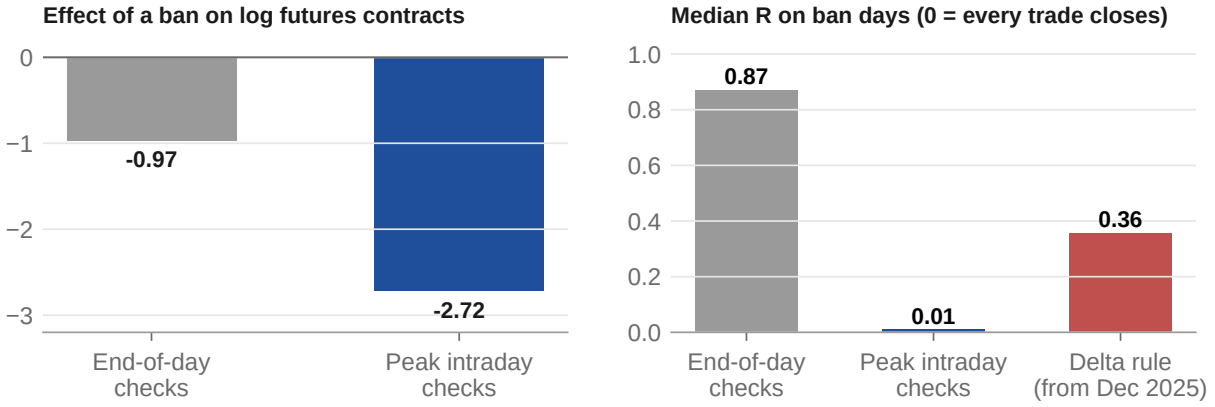
OUTCOME	EFFECT (S.E.)	NON-BAN MEAN
log(1 + futures contracts traded)	−2.00 (0.11)	
log(1 + cash shares traded)	+0.073 (0.027)	
Delivery share of cash volume	+0.041 (0.005)	0.26

Stock-by-utilisation-bin and date fixed effects; standard errors clustered by stock and date; expiry days excluded. Source: author's calculations.

FIGURE 7.1

The ban bites as designed only when positions are checked during the day

Effect of a ban on log futures contracts (left) and median *R* on ban days (right), by how violations were judged



End-of-day checks to 22 March 2020; peak intraday checks from 23 March 2020 to September 2025; delta rule from October 2025 (from 8 December 2025 shown). Source: author's calculations.

prices, gives −0.190 (s.e. 0.077) for the upper regime and +0.088 (0.015) for the lower: the result is not caused by noisier prices on ban days (Figure 7.3).

A reading consistent with these facts, though not tested in advance, is that at the start of a ban the model's second channel dominates: demand for new futures exposure is blocked, which removes the pressure that had made futures rich, so upward deviations close quickly. The prediction tested was derived from the reduce-only arbitrage channel alone.

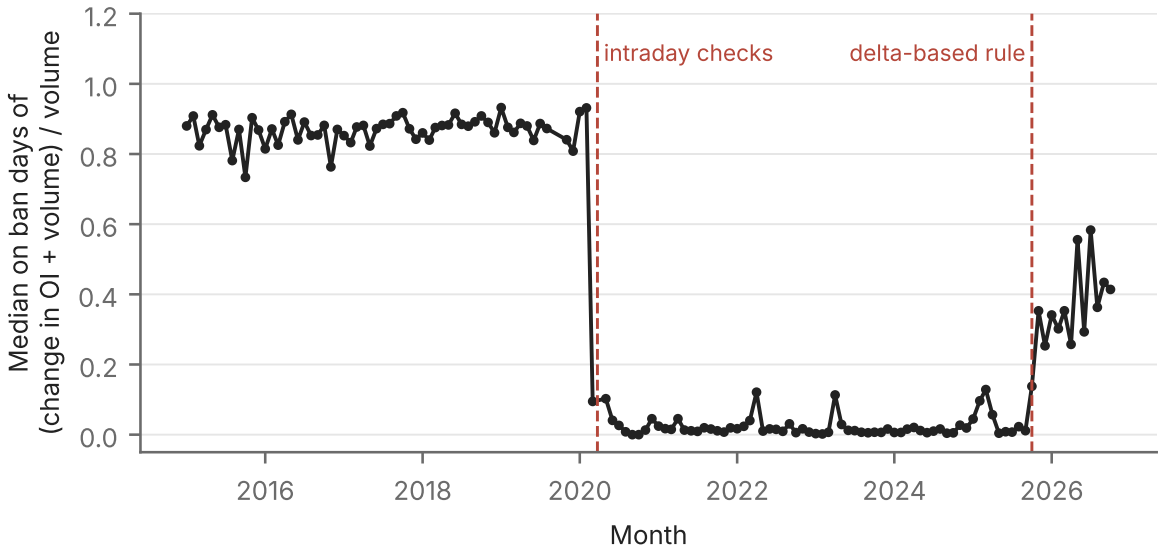
7.3. THE LEVEL OF THE BASIS WHEN A BAN STARTS

Design. Each ban spell from 2015 to September 2025 with no ban in the five days before it is an event. Controls are other stocks in the same week with utilisation between 85% and 95% and no ban within five days, matched (up to five, nearest by Mahalanobis distance) on the utilisation path of the previous five days and the basis on the day before. The outcome is the change in the basis from the day before the ban, treated minus matched controls, averaged over the ban days

FIGURE 7.2

When intraday checks began in March 2020, every futures trade in a ban started to close positions

Monthly median of $R = (\Delta OI + V)/V$ on ban days, near and next futures, expiry days excluded



$R = 0$ when every trade closes a long and a short; $R \approx 1$ on normal days. Source: NSE F&O bhavcopies and ban lists; author's calculations.

TABLE 7.2

In a ban rich futures revert faster, not slower

Slope of the daily change in the basis on yesterday's deviation, by regime, and its change on ban days

	NORMAL SLOPE (S.E.)	CHANGE IN A BAN (S.E.)
Upper regime	−0.712 (0.010)	−0.089 (0.037)
Middle	−0.393 (0.026)	+0.099 (0.062)
Lower regime	−0.204 (0.010)	+0.081 (0.030)

Regimes from each stock's 20th and 80th percentiles on non-ban days; stock and date fixed effects; standard errors clustered by stock and date. Source: author's calculations.

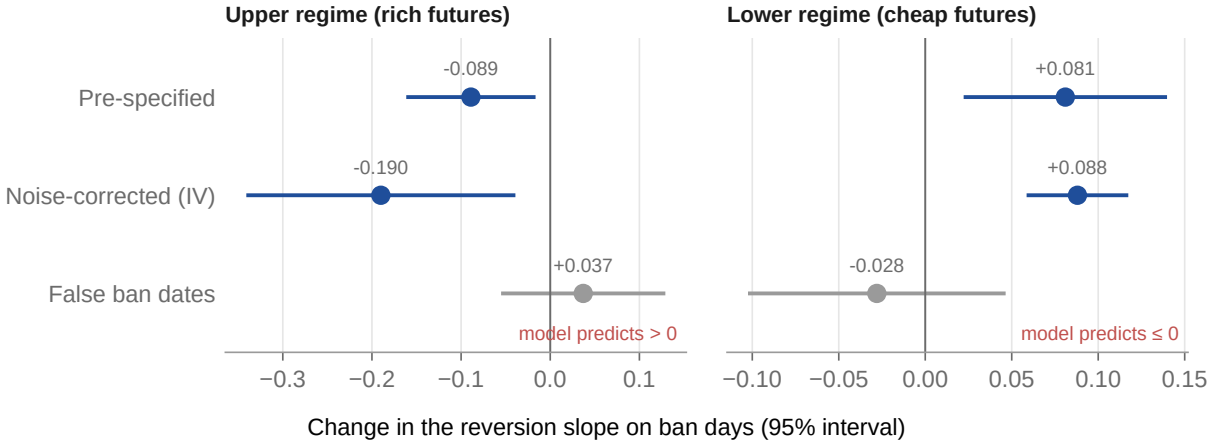
among the first five. Composition C is the change in futures open interest over the five days before entry divided by futures volume.

Results. Over 889 matched entries the basis falls by 6.3 bp on average (median −7.8). With falling open interest before entry ($C \leq 0$, 30 events) it rises by 8.4 bp (s.e. 10.9); with rising open interest it falls by 6.8 bp (s.e. 1.8). The difference, the pre-specified statistic, is −15.2 bp (one-sided $p = 0.083$): the signs are those of the model's level result, but the closing-pressure group is small because open interest must rise for utilisation to cross 95%. Splitting at the median of C instead gives a difference of −5.8 (s.e. 2.9, $p = 0.025$), and C as a continuous variable gives −4.8 per standard deviation (s.e. 1.6, $p = 0.001$). The basis returns to its earlier level about a week after the start (Figure 7.4).

FIGURE 7.3

The opposite-sign result survives a noise correction and vanishes at false ban dates

Change in the reversion slope on ban days: pre-specified estimate, instrumented by the deviation two days earlier, and at false ban dates



False ban dates: every spell before October 2025 moved 60 trading days later. Source: author's calculations.

Caveats. The matching leaves the treated basis 10 bp below that of the controls on the day before, and the treated basis dips on that day; with the baseline taken over days −10 to −2, or controlling for the gap, the in-ban fall is 9 to 11 bp and the composition slope stays negative. With controls from the same day only (685 events) the fall is −4.0 bp (s.e. 2.0) but the composition gradient is not significant. Classifying events instead by the direction of open interest and price (long build-up, short build-up, short covering, long unwinding), the in-ban fall appears for both long build-up (−7.9 bp) and short build-up (−5.5 bp), so it is not specific to new long demand.

7.4. THE 95% TRIGGER

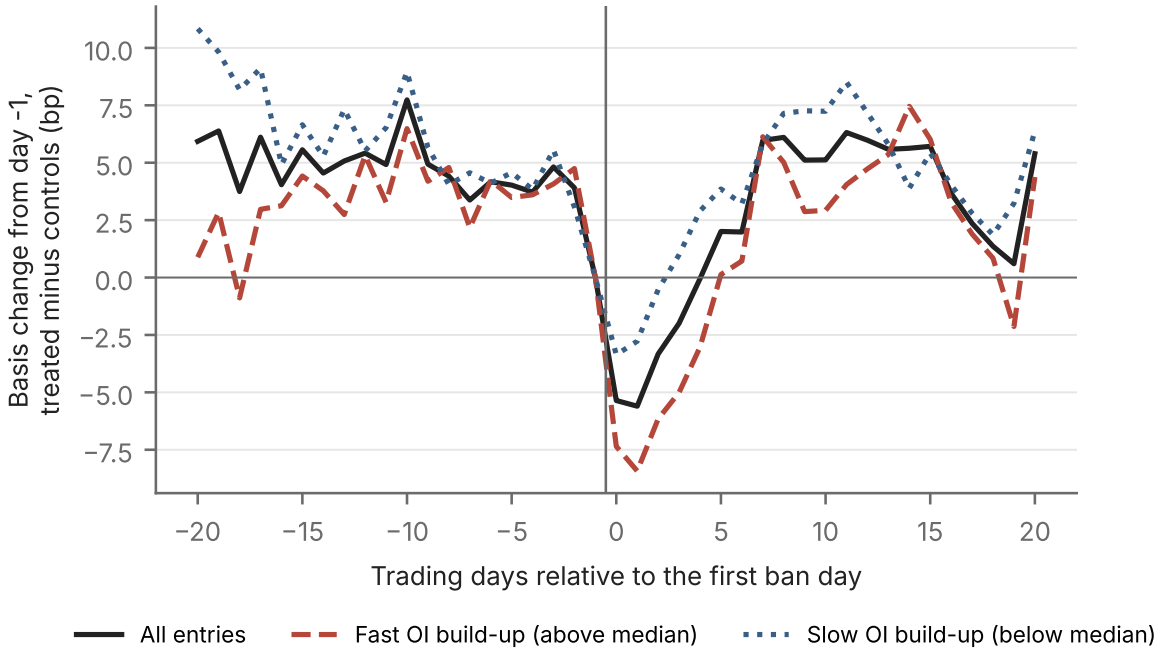
Density. Among stocks that were below 95% the day before, the density of today's utilisation is smooth through the trigger (Figure 7.6; Cattaneo, Jansson and Ma test, $T = -0.18$, $p = 0.86$), across polynomial orders, bandwidths and enforcement periods. Utilisation accelerates rather than stalls near the trigger: its mean daily change is larger just below 95%, and the share of stocks crossing the next day rises steadily to about one half at 0.93 to 0.95. There is no sign that traders hold back to avoid a ban.

Regression discontinuity. For stocks not in a ban the day before, crossing 95% puts the stock in the ban the next day in 99.7% of cases. Local linear estimates with robust bias-corrected inference show sharp jumps in trading: futures contracts −1.53 log points, R −0.66, delivery share +0.051; cash volume does not jump. The basis level is 12.9 bp lower above the cutoff ($p = 0.02$), but the basis is already 11.3 bp lower on the crossing day itself, before the ban starts, and the change on the first ban day is small (−2.8, s.e. 4.1). The discontinuity therefore confirms the trading mechanism but cannot separate a level effect of the ban from the pressure that pushes utilisation over the trigger (Figure 7.7).

FIGURE 7.4

The basis falls when a ban begins, more after fast build-ups of open interest, and recovers within about a week

Change in the basis from the day before a ban starts, treated minus matched controls, by event time (889 entries, 2015 to September 2025)



Controls: other stocks in the same week at 85% to 95% utilisation, matched on the utilisation path and the basis. Source: author's calculations.

Hysteresis. Between 80% and 95% utilisation, stocks in a ban and stocks not in a ban can be compared at equal utilisation. Controlling for utilisation, its change, five lags of the basis and stock and date fixed effects, being in a ban changes the basis by +0.4 bp (s.e. 0.8, $p = 0.64$); without the lags the difference is −14.1 bp. Banned stocks carry the low basis of the start of the spell; the reduce-only state does not shift it further. Its effect on trading is as strong as in H8.

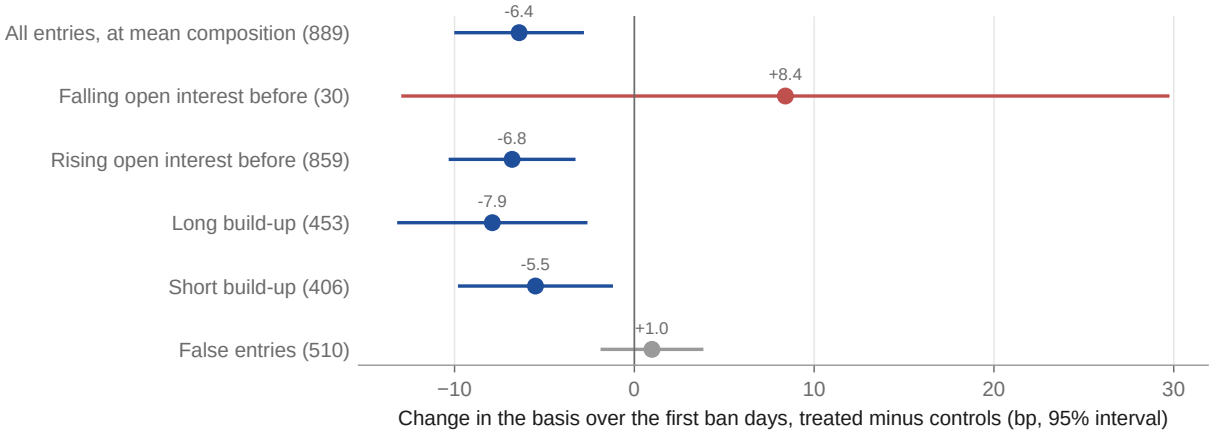
7.5. BAD NEWS IN A BAN (H2)

Of 8,586 results announcements of F&O stocks, 179 fall on ban days, 55 of them bad news, so this test has little power, as stated in advance. The drift after bad news is −1.25% outside bans relative to the middle tercile; in bans the additional effect is +0.7% (s.e. 3.4). Outside bans the basis over the five days before an announcement is 2.0 bp lower before bad news than before good news (s.e. 1.1, one-sided $p = 0.04$), the pattern reported by Saurav, Srivastava and Agarwalla (2026); inside bans the difference cannot be estimated with useful precision.

FIGURE 7.5

The fall appears for both long and short build-ups and not at false entries

Mean change in the basis over the first ban days, treated minus matched controls, by group (events in brackets)

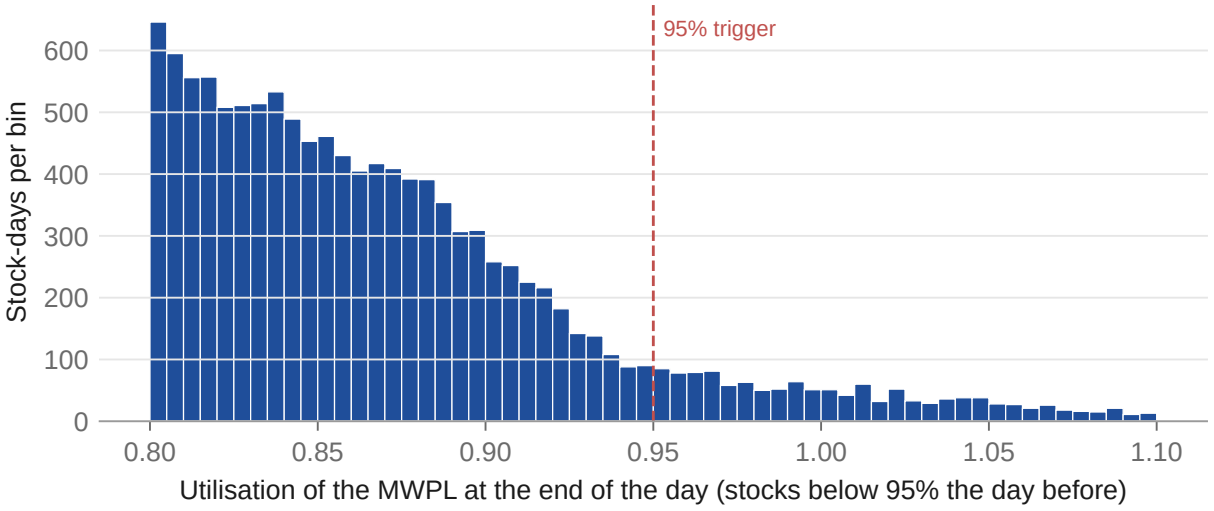


Source: author's calculations.

FIGURE 7.6

Utilisation does not bunch below the trigger

End-of-day utilisation of stocks that were below 95% the day before, 2015 to September 2025



Cattaneo, Jansson and Ma test: $T = -0.18$, $p = 0.86$. Source: NSE combined open interest files; author's calculations.

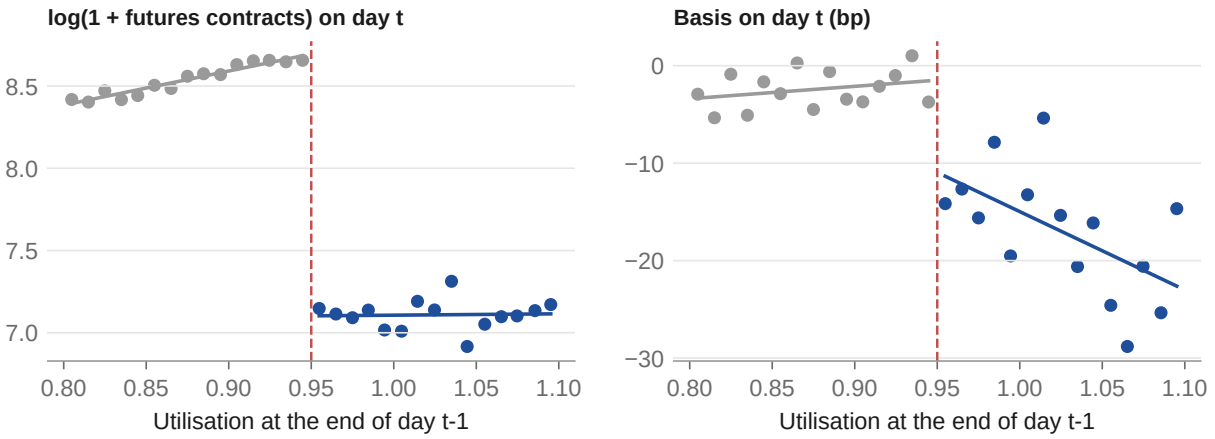
7.6. PLACEBO CHECKS

Two of the results above could in principle come from the method rather than from bans: the regime slopes of H7 and the matched-control design of the level effect. I therefore move every ban spell before October 2025 sixty trading days later, for the same stock and with the same length, keeping only the 871 false spells that do not touch a real ban, and rerun both designs with the false spells in place of the real ones. The H7 changes become +0.037 (s.e. 0.047) for the upper regime and −0.028 (0.038) for the lower, against −0.089 and +0.081 for real bans. In the level-effect design the basis changes by +1.0 bp at false entries, against −6.3 at real ones, with no gradient in the composition of open interest (difference −0.3 bp, s.e. 3.0). Neither result

FIGURE 7.7

Futures trading drops sharply at the trigger; the basis is lower above it

Binned means by utilisation at the end of the previous day, stocks not in a ban the day before, with linear fits on each side



The estimates in the text are local linear with robust bias-corrected inference. Source: author's calculations.

is produced by the method.

7.7. SUMMARY

Bans bind hard on trading, and the identity that every trade closes positions holds once the rule is enforced on intraday positions. The basis does not behave as the reduce-only arbitrage channel alone predicts: rich futures revert faster, and the basis falls when a ban starts, more after fast build-ups of open interest. Both facts fit the model's second channel, blocked new demand, but that reading was formed after seeing the results. Neither result appears at false ban dates.

CHAPTER 8

F&O eligibility and the cross-section

Short-sale constraints across the whole cash market: what entry into and exit from the F&O segment do to price adjustment and drift, whether anomalies are stronger where shorting is hardest, and how drift and price delay vary across all stocks.

8.1. ENTRY INTO AND EXIT FROM THE F&O SEGMENT (H6)

Entry gives a stock futures, options and formal short-selling eligibility at once; an eligibility exit removes them while the stock keeps trading in the cash market. This changes the short-sale constraint for many more stocks than the ban does, and against the whole cash market as a control group.

Design. Stocks by calendar year, 2015 to 2025. A stock's cohort is the first year fully after its event; the event year is a transition year. For price delay I use the Callaway and Sant'Anna (2021) estimator with never-treated and not-yet-treated controls and base-year covariates (log market capitalisation, log traded value, volatility), with a stock-cluster bootstrap. Drift exists only at announcements, so for drift after bad news I use a stacked difference-in-differences at the announcement level. The family statistic is a Bonferroni combination of the four components.

Prices of stocks that enter the segment respond much faster to market-wide news (median delay 0.24 before entry), and those of stocks that leave somewhat more slowly. The Sun and Abraham (2021) estimator gives -0.063 (s.e. 0.019) for entries, leads three and four years before entry are close to zero, and the result holds without the large November 2024 batch and without back-cast market capitalisations. The drift components go the other way: after entry the drift that follows bad news is stronger, not weaker. H6 is supported through price delay and contradicted through drift. Entry requires a stock to have grown in size and liquidity, which by itself speeds price adjustment; the covariates and the flat leads limit, but cannot rule out, selection on growth (Figure 8.1).

8.2. ANOMALIES AND SHORT-SALE CONSTRAINTS (H5)

Each month I sort stocks within three constraint groups (F&O stocks; cash-only stocks with SLB trades in the last three months; cash-only stocks without) into terciles of a composite mispricing score built from momentum, short-term reversal, maximum daily return, idiosyncratic volatility, illiquidity and turnover, and measure the most overpriced and most underpriced terciles against

TABLE 8.1

Entry speeds price adjustment; drift after bad news does not weaken

Pre-specified components of the H6 test

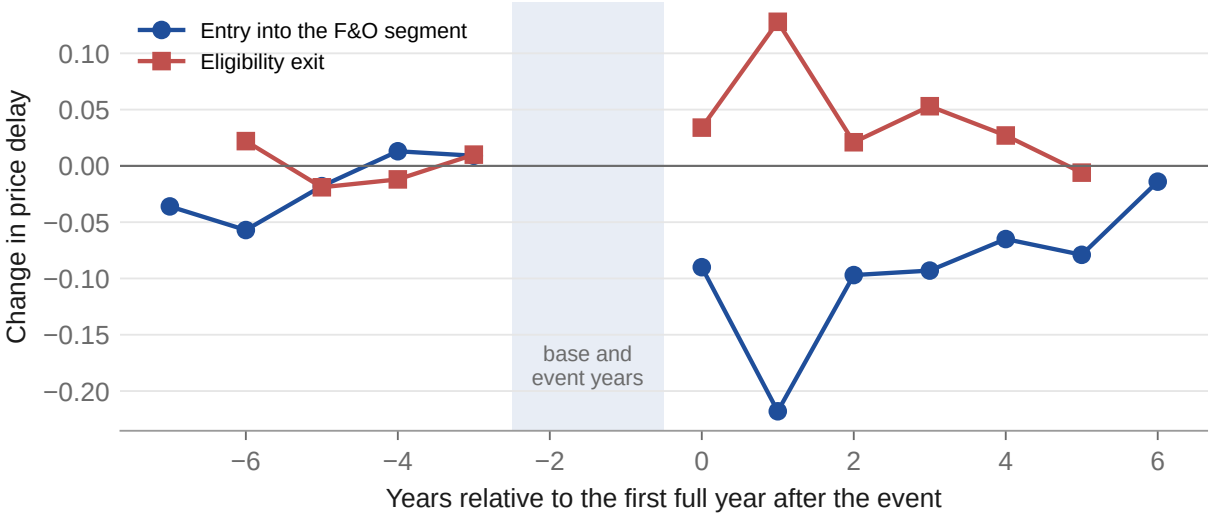
COMPONENT	PREDICTION	ESTIMATE	S.E.	ONE-SIDED p
Entry: price delay	< 0	-0.109	0.023	1.3×10^{-6}
Entry: drift after bad news	> 0	-0.040	0.018	0.99
Exit: price delay	> 0	$+0.053$	0.040	0.091
Exit: drift after bad news	< 0	$+0.014$	0.020	0.77

Price delay: Callaway and Sant'Anna estimator, stock-cluster bootstrap. Drift: stacked difference-in-differences at the announcement level. Source: author's calculations.

FIGURE 8.1

Price delay falls after entry and rises a little after exit, with no clear trend before either

Callaway and Sant'Anna estimates of the change in price delay by year relative to the first full year after the event (cells with at least 19 treated stocks)



Point estimates; the aggregate post-event effects and their standard errors are in Table 8.1. Source: author's calculations.

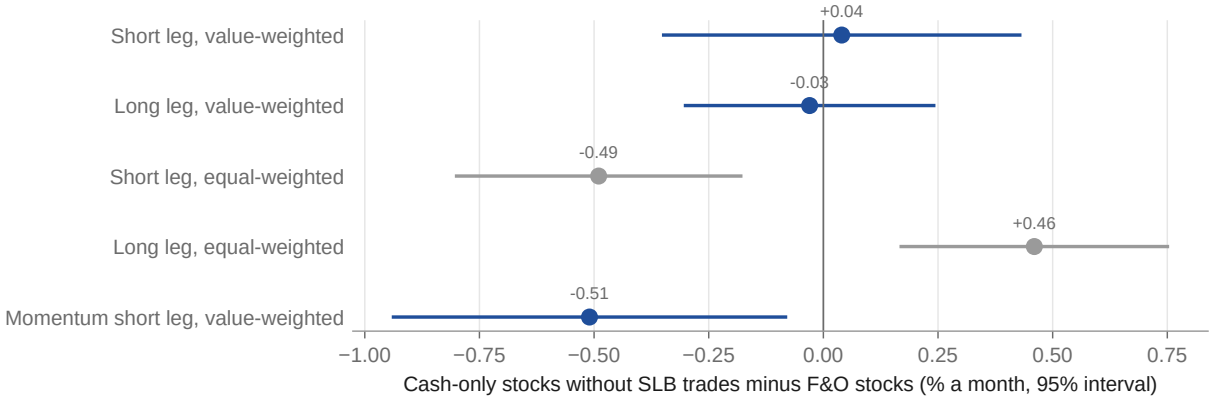
the group's value-weighted return. The model predicts that the overpriced side does worse where shorting is hardest, while the underpriced side does not differ.

Value-weighted, it does not: the short-leg difference between the most constrained group and F&O stocks is +0.04% a month (s.e. 0.20). Equal-weighted, anomalies are far stronger among small cash-only stocks, but on both sides (short leg −0.49%, long leg +0.46% a month relative to F&O stocks), which is a small-stock effect rather than the one-sided pattern of a short-sale constraint. Momentum alone fits the prediction: losers among constrained stocks do worse (−0.51% a month, s.e. 0.22). The groups differ greatly in size (median market capitalisation Rs 43,513 crore for F&O stocks against Rs 770 crore for cash-only stocks without SLB trades), which limits what this comparison can show (Figure 8.2).

FIGURE 8.2

Value-weighted, the overpriced side does no worse where shorting is hardest

Most constrained group (cash-only stocks without SLB trades) minus F&O stocks, short and long legs of a composite mispricing sort



Newey-West standard errors; holding months January 2016 to September 2026. Source: author's calculations.

8.3. DRIFT AND PRICE DELAY ACROSS ALL STOCKS

Two further comparisons use the whole cash market. First, relative to F&O stocks of the same size decile, cash-only stocks drift −0.67% more after bad news (s.e. 0.42) and +2.40% more after good news (s.e. 0.43): they under-react in both directions. Second, the constraint index of Chapter 5, which combines the lack of a short channel with SLB scarcity, spreads, illiquidity and idiosyncratic volatility, predicts slower price adjustment the next year, also within a stock over time and after controlling for size. It also predicts more drift after bad news than after good news, but that asymmetry disappears once size is controlled for, in line with the symmetric under-reaction of cash-only stocks.

8.4. SUMMARY

Short-sale access matters for how fast prices adjust: entry into the F&O segment speeds the response to market news, and more constrained stocks adjust more slowly. The evidence on overpricing and on drift after bad news is mixed: entry does not weaken bad-news drift, value-weighted anomaly returns do not show the predicted asymmetry, and the link between the broader constraint index and drift after bad news disappears once size is controlled for.

CHAPTER 9

Conclusion and policy

What the tests say about which limits bind, what the results imply for the regulation of position limits and short selling, and what the data cannot show.

9.1. WHAT BINDS

The answer to the question this thesis asks is uneven across the limits. Position limits bind hard on trading: the ban moves trading out of futures as soon as it is enforced on intraday positions. Their effect on prices is real but not the one the reduce-only arbitrage channel predicts; the evidence points instead to the blocking of new demand at the start of a ban. Short-sale access matters for how fast prices adjust to market news, both when a stock enters the F&O segment and across all stocks, but less clearly for overpricing. Trading costs shape the band, which is wide and asymmetric, but its edges do not move one-for-one with the tax: the full cost of a new position lies in the outer tail of the basis, and the cost of keeping arbitrage capital in place is small. Arbitrage capital leaves weak traces in the predicted direction.

With ten secondary statistics reported without adjustment, one or two values below 0.05 would be expected by chance. The secondary results should be read as a pattern, not one by one.

9.2. POLICY

Position limits and the ban exist to contain concentrated positions and manipulation in a market where speculative demand is large. SEBI's study of individual traders in equity derivatives (SEBI, 2024a) shows how large: over the three years to March 2024, 93% of more than one crore individual traders lost money, on average about Rs 2 lakh each including transaction costs, and their aggregate losses exceeded Rs 1.8 lakh crore. In the last of those years proprietary traders and foreign portfolio investors earned gross profits, 96 to 97% of them from algorithmic trading. SEBI's measures of October 2024 for index derivatives cite the rise in retail participation and speculative volumes (SEBI, 2024b), and its measures of May 2025 changed how open interest is measured and how position limits are set (SEBI, 2025). Most of that activity is in index options, not in the single-stock futures studied here, but the same pressure of new speculative demand is what pushes a stock towards its limit.

The results bear on the trade-off in four ways. First, the ban is costly for liquidity: futures trading falls by almost 90% in a ban, while cash trading rises a little. Second, its main price effect

TABLE 9.1

Three of four primary hypotheses are supported on the pre-specified statistic; one is rejected with the opposite sign

All pre-specified tests, with the result and the main qualification

HYPOTHESIS	PRE-SPECIFIED RESULT	COMMENT
PRIMARY FAMILY (HOLM CORRECTION AT 5%)		
H8: trading leaves futures in a ban	Supported ($p < 10^{-10}$)	Identity holds under intraday enforcement
H3: costs set the band edges	Supported ($p < 10^{-10}$)	Placebo-calibrated evidence weak; mostly a common shift
H6: short-sale access	Supported ($p = 5 \times 10^{-6}$)	Through price delay; drift goes the other way
H7: one-sided correction	Rejected, opposite sign	Survives a noise correction
SECONDARY (UNADJUSTED)		
Level effect by composition	$p = 0.083$	Signs as predicted; median split $p = 0.025$
Density at 95%	No discontinuity	No bunching
Regression discontinuity	Basis −12.9 bp, $p = 0.02$	Gap present before the ban
Hysteresis	+0.4 bp, $p = 0.64$	No extra level shift
H9 anticipation	$p = 0.019$	Upper tail grows toward 95%
H9 reform	$p = 0.11$	Right direction
H2 bad news in bans	$p = 0.59$	No power
H1 frictions and band	$p = 0.002$	Across stocks only
H4 capital	$p = 0.11$	Clearer in differences
H5 anomalies	$p = 0.57$	Not supported value-weighted

Source: Chapters 6 to 8.

appears at the start of a ban, where it fits the blocking of new demand, and fades within about a week; once a stock is in the ban, staying in it does not move the basis further. If that reading is right, a ban restrains the demand it is designed to restrain, but at the cost of the arbitrage that would otherwise keep futures and cash prices together. Third, enforcement matters: the rule changes trading in the way it is meant to only when positions are checked during the day. The October 2025 rule, which allows any trade that lowers an entity's end-of-day delta, has visibly loosened the link between volume and open interest in banned stocks.

Fourth, the 2025 reform met its stated aim of fewer “spurious” bans (SEBI, 2025, para. 4.2). Before the reform, bans were not caused by falls in the limit: almost every entry came from rising open interest. What made many of them avoidable was the measure of open interest, which counted every option at its full notional value. Measuring positions in delta terms removes almost all of these bans, the new and lower limits add some back, and the combination cut ban days by about three quarters in the first year. Whether the new formula has changed the cost of each ban needs more post-reform data than the 44 spells that began after the reform. The 2020 restoration of the limit adds one more lesson: a looser limit lowered utilisation and bans and made a rich basis less common, so the level of the limit affects prices even before a ban starts.

The SLB market remains thin. Because only F&O stocks may be sold short and the overnight short is in practice a stock future, entry into and exit from the F&O segment change the short-sale constraint of a stock at once. The faster price adjustment after entry is one measurable benefit of that access.

9.3. LIMITATIONS AND FURTHER WORK

The data are daily, so intraday arbitrage and the exact timing of trades within the day are outside the scope, and noise in daily prices inflates measured reversion. Arbitrage inventory is not observed. The post-reform period is short. Three papers relevant to positioning (Saurav, Srivastava and Agarwalla, 2026; Gagnon, 2018; Brennan and Schwartz, 1990) could not be read in full. Intraday data would allow sharper tests of the ban’s effect on correction, and a longer post-reform sample would allow a proper test of the 2025 formula.

APPENDIX A

The pre-analysis plan and its log

A.1. THE PLAN

Before any outcome (basis, returns, drift, delay) was related to any treatment (ban status, utilisation, MWPL changes, STT dates, F&O entry or exit, the tax change), I wrote a pre-analysis plan fixing the samples, the outcome and treatment definitions, the specifications, the predicted signs, the falsifiers, the inference and the primary and secondary hypotheses. It was frozen with a time-stamped tag in the project's version control on 5 November 2026, after the guide's go-ahead and a full set of checks on the data. Until then, only treatment-side counts and outcome distributions on their own had been examined.

- *Primary family:* H7 (one-sided correction), H8 (trading), H3 (costs and band edges), H6 (F&O entry and exit), corrected jointly by the Holm method at 5%.
- *Secondary tests:* the level effect, H9, the regression discontinuity at 95% with a density test, hysteresis, H2, H1, H4 and H5, reported without adjustment and labelled exploratory.
- *Inference:* standard errors clustered by stock and date; bootstrap where clusters are few; winsorisation at 1% and 99% within each year, with robustness without.

A.2. THE LOG

Where the plan left a detail open (for example the exact sample filters, the composition split for the level effect, the population for the density test, or the outcome window for drift), I wrote the detail into a dated log, committed it, and only then ran the test. Analyses added after seeing a result are marked as exploratory in the text and listed in the log; the pre-specified result is always reported first. The main entries were:

- the analysis sample, the H7 regimes and test statistic, the H8 comparison zone and identity reading, the H3 windows and placebo, and the H6 design (calendar-year cohorts, controls, covariates, outcome definitions, stacked design for drift, a Rs 5 price filter for event outcomes), all before the primary tests;
- before each secondary test, its design: event and control definitions and a median split for the level effect (only 8% of entries have falling open interest before entry); a density test on stocks below the trigger the day before (because the density over all non-ban days thins above the trigger by construction); pre-determined placebo outcomes for the discontinuity;

five basis lags for hysteresis; stock-specific tail measures and the reform intensity for H9; the interaction design for H2; a principal-component friction index for H1; lagged flows with tax and STT dummies for H4; constraint groups for H5;

- before the band estimation and the constraint index, their specifications;
- exploratory additions: the enforcement-period split for H8, diagnostics and a noise correction for H7, the placebo calibration of H3, an alternative baseline and a build-up classification for the level effect, bandwidth checks for the density test, period-specific demeaning and a noise check for the band.

APPENDIX B

Data construction

B.1. BAN AND POSITION-LIMIT PANEL

Daily combined open interest files give the MWPL, open interest, futures-equivalent open interest (from October 2025) and the next-day status of each stock from February 2010. About a hundred files write the year in two digits, a few carry another day's data under their own date, and some archives bundle several days; the parser takes dates from the file names, reassigns misdated files and reads every member of bundled archives. Thirteen files from June 2012 are corrupt at the source (before the main sample). Placeholder symbols in some files are restored through the security's identifier (ISIN). The official ban lists are parsed with their trailing punctuation removed; on three days the list is missing at the exchange and is imputed.

B.2. F&O MEMBERSHIP

A stock is in the F&O segment on a day if stock futures on it are listed that day. Entries and exits are the first and last days of each membership spell; symbol changes are linked through the ISIN so that they are not counted as events. The F&O file of 30 March 2021 is missing at the exchange and is bridged.

B.3. PRICES AND THE BASIS

Futures and cash prices, traded value and quantity, contracts, open interest and delivery quantities come from the daily bhavcopies, in the older and the newer (UDiFF, from July 2024) formats. The older futures files do not report lot sizes; they are recovered from open interest and checked against the day's price range (the futures average price lies inside the day's high and low for 99.8% of contracts). The futures volume-weighted price is traded value divided by contracts times lot size; the cash price is traded value divided by quantity. Dividends going ex before expiry are subtracted at their present value; contracts with a dividend or another corporate event in their life are flagged and excluded from the main sample, because adjusting with actual ex-dates over-corrects slightly (some dividends were not yet announced). The rate is the three-month interbank rate; the last published value is carried forward at most two months and flagged.

B.4. RETURNS AND MARKET CAPITALISATION

Daily returns for all stocks use the close and the exchange's previous close, which is not adjusted for corporate actions; splits and bonuses (multiplied when they fall on the same day) and dividends are applied from the corporate-actions feed, missing splits are inferred from price ratios at standard split ratios, and listing days and other event days are set to missing. Market capitalisation comes from AMFI's half-yearly average market-capitalisation lists (from July 2017), turned into share counts and applied to split-adjusted prices; before July 2017 share counts are carried back.

B.5. RESULTS ANNOUNCEMENTS

The release time of each quarterly result is the earliest of the exchange's "financial result updates" announcement and the results filing; board-meeting dates fill gaps and replace filing times that are more than a day after the meeting; each record is mapped to the symbol trading on the event date through the ISIN history. Day 0 is the release day for releases before 15:30 and the next trading day otherwise.

B.6. SECURITIES LENDING AND ARBITRAGE FUNDS

The daily SLB files list each lending trade's stock, tenure, expiry, fee, quantity and value (2015 to 2026; two days missing at the exchange). AMFI's monthly reports give the arbitrage-fund category's flows and assets from April 2018, in two layouts.

B.7. REPRODUCIBILITY

Every data set is rebuilt from the raw files by scripts, each with a report of counts and checks; the repository's documentation lists the sources, the construction rules, the checks run on the data and the exact rebuild sequence. From the processed data, every estimate in this thesis is produced by one sequence of analysis scripts; a complete rerun reproduced every reported number. Each estimator was first run on data simulated with a known effect (the threshold model, the Callaway and Sant'Anna estimator and the matched level-effect design all recover the effect they were given), and a separate check compares every number quoted in the text with the analysis output.

APPENDIX C

The model in full

The full model behind Chapter 4: the environment, the band, inventory, the ban as a reduce-only rule, anticipation, identification, short-sale constraints beyond the basis, the estimation lens, calibration, the map from results to hypotheses, and the numerical checks.

C.1. PURPOSE AND DESIGN RULES

This appendix gives the full model summarised in Chapter 4, with proofs and numerical checks. The aim was to fix a theoretical framework before the empirical work, so that the hypotheses are derived rather than fitted. The model builds on the limits-to-arbitrage literature (Shleifer and Vishny, 1997; Gromb and Vayanos, 2010) and on the econometrics of no-arbitrage bands (Dwyer, Locke and Yu, 1996; Tse, 2001; Ters and Urban, 2020). Three design rules guide it:

1. **Institutions as they are.** The ban, the short-selling framework, settlement prices and costs are modelled from the primary sources (SEBI circular 2025/79; NSE Clearing circular 119/2025 and its FAQ of October 2025).
2. **Predictions that can fail.** Wherever the sign of an effect depends on the environment, the model says on what, and the empirical work tests that condition rather than a one-sided sign.
3. **Daily horizon.** The data are daily, so the relevant arbitrageur carries positions overnight (arbitrage mutual funds, proprietary desks), not a one-second market maker.

C.2. ENVIRONMENT

C.2.1 Prices and the basis

A stock with spot price S_t has a single-stock future maturing at T. With benchmark rate r_b (e.g. MIBOR) and known dividends, the cost-of-carry value is $F_t^* = (S_t - \mathrm{PV}_t(D))\, e^{r_b(T-t)}$. The *basis* (mispricing) is

$$m_t \;=\; \ln F_t - \ln F_t^*, \tag{C.1}$$

measured in basis points of the spot price. Write s and f for the deviations of spot and futures prices from the fundamental value, so $m = f - s$ to first order.

C.2.2 Outside traders and market clearing

Outside (non-arbitrage) traders in each market have linear demand. In the cash market net demand is $D_S - \psi_S s$, in the futures market net long demand is $D_F - \psi_F f$; $\psi_S, \psi_F > 0$ measure liquidity (the depth of the outside side of each market) and D_S, D_F are demand shifters. An arbitrage sector holds a units of the cash-and-carry trade: long a shares and short a futures ($a < 0$ is reverse arbitrage). Both markets are in zero net supply after removing the fundamental holdings, so

$$D_S - \psi_S s + a = 0, \qquad D_F - \psi_F f - a = 0. \tag{C.2}$$

Subtracting, the basis satisfies

$$m = \frac{u - a}{\Phi}, \qquad \frac{1}{\Phi} \equiv \frac{1}{\psi_F} + \frac{1}{\psi_S}, \qquad u \equiv \Phi\Big(\frac{D_F}{\psi_F} - \frac{D_S}{\psi_S}\Big). \tag{C.3}$$

u is the *relative demand pressure* toward futures: positive when outside traders want long exposure in futures more than in cash (e.g. leveraged speculative longs), negative when they want short exposure in futures (hedgers, informed pessimists who can short only through futures). Without arbitrage the basis is u/Φ; arbitrage lowers it by a/Φ (buying cash raises s, selling futures lowers f).

C.2.3 The arbitrage sector

Assumption C.1 (Arbitrage)**.** The arbitrage sector is competitive and chooses a to maximise

$$\Pi(a) = a\,m - \tfrac{\gamma}{2}a^2 - c_+\,a^+ - c_-\,a^-, \qquad \gamma \equiv \alpha\sigma^2 > 0, \tag{C.4}$$

where $a^{\pm} = \max(\pm a, 0)$, α is risk aversion per unit of capital (inversely related to arbitrage capital), σ^2 the basis risk borne until convergence (margin calls, early unwinds), c_+ the cost of a cash-and-carry position and c_- the cost of a reverse position.

Holding a unit of cash-and-carry to expiry earns m (the basis converges to zero), less costs. Decomposing,

$$c_+ = \tau_+ + (r_A - r_b)(T - t), \qquad c_- = \tau_- + \big(\ell_s + r_b - r_L\big)(T - t), \tag{C.5}$$

with $\tau_{\pm}$ the round-trip transaction costs of each trade (STT, stamp duty, exchange fees, brokerage, impact), r_A the required return of cash-and-carry capital, ℓ_s the stock-lending fee and r_L the rebate on short proceeds. Section C.11 maps each Indian limit to these parameters.

C.2.4 Indian limits as parameters

LIMIT	INDIAN INSTITUTION (DATED SHOCKS)	PARAMETER
Trading costs	STT on futures (sell side): 0.01% from 1 Jun 2013; 0.0125% from 1 Apr 2023; 0.02% from 1 Oct 2024; 0.05% from 1 Apr 2026. Cash delivery STT 0.1% each side; 0.1% on physically settled derivatives	τ_+, τ_-
Short-sale cost	Naked shorting barred; only F&O-segment securities are eligible for short selling; SLB (which also covers other liquid stocks) thin; the overnight short is the future	c_- via ℓ_s; $c_- = \infty$ for cash-only stocks
Arbitrage capital	Arbitrage-fund AUM (tax-advantaged since the 1 Apr 2023 debt-fund tax change); funding rates; peak-margin phase-in (Dec 2020–Sep 2021)	α (so γ), r_A
Arbitrage risk	Basis volatility, idiosyncratic volatility	σ^2 (so γ)
Position limits	Entity limits (10/20/30% of MWPL from 1 Oct 2025); MWPL; the F&O ban (95% entry, 80% exit)	caps on a; reduce-only rule (Sec. C.5)
Market depth	Liquidity of outside traders in each market	ψ_S, ψ_F (so Φ)

C.3. THE NO-ARBITRAGE BAND

Proposition C.1 (Asymmetric band and passthrough)**.** *Under Assumption C.1 and* (C.3)*, the equilibrium is unique:*

$$m(u) = \begin{cases} \dfrac{\gamma u + c_+}{\gamma\Phi + 1}, & u > \Phi c_+, \\ \dfrac{u}{\Phi}, & -\Phi c_- \le u \le \Phi c_+, \\ \dfrac{\gamma u - c_-}{\gamma\Phi + 1}, & u < -\Phi c_-, \end{cases} \qquad a(u) = \begin{cases} \dfrac{u - \Phi c_+}{\gamma\Phi + 1}, & u > \Phi c_+, \\ 0, & \textit{inside}, \\ \dfrac{u + \Phi c_-}{\gamma\Phi + 1}, & u < -\Phi c_-. \end{cases} \tag{C.6}$$

The basis is left alone inside the band $[-c_-, c_+]$*; outside it, arbitrage absorbs a share of each demand shock, so the passthrough falls from* $1/\Phi$ *to* $\gamma/(\gamma\Phi + 1)$*.*

Proof. Π is strictly concave with kinks at 0. The best response is $a = (m - c_+)/\gamma$ if $m > c_+$, $a = (m + c_-)/\gamma$ if $m < -c_-$, and $a = 0$ otherwise. The map $a \mapsto \mathrm{BR}((u - a)/\Phi) - a$ is strictly decreasing, so the fixed point is unique. Substituting $m = (u-a)/\Phi$ into the interior best responses gives (C.6); e.g. for the upper branch $m\Phi = u - (m - c_+)/\gamma$, i.e. $m = (\gamma u + c_+)/(\gamma\Phi + 1)$, which equals c_+ at $u = \Phi c_+$ (continuity). □

Proposition C.2 (Comparative statics)**.** *(i)* Costs set the edges. *The band edges are c_+ and $-c_-$; they do not depend on capital or risk. A change in transaction costs $\Delta\tau_\pm$ (e.g. an STT change) shifts the corresponding edge one-for-one.*

(ii) Capital and risk set the passthrough. *Outside the band, $\partial m/\partial u = \gamma/(\gamma\Phi + 1)$ rises with $\gamma = \alpha\sigma^2$: scarcer capital or riskier convergence lets the basis travel further beyond the edge.*

(iii) Maturity. *By* (C.5) *both edges are affine in remaining maturity $T - t$ and converge to the round-trip costs τ_+, $-\tau_-$ at expiry (for positions held to expiry).*

(iv) Location. *The band is centred at $(c_+ - c_-)/2$, below zero whenever shorting costs more than carrying ($c_- > c_+$). Its* upper *edge is itself negative when $(r_b - r_A)(T - t) > \tau_+$, i.e. when cash-and-carry capital accepts a return sufficiently below the benchmark rate.*

Proof. (i)–(iii) follow from (C.6) and (C.5). (iv): the centre is $(c_+ - c_-)/2$; the sign of c_+ follows from (C.5). □

Remark. Part (iv) matters because the Indian evidence shows a band centred below zero: Shankar, Sankar and Kiran Kumar (2019) estimate mean thresholds of -39 and -10 bp for 102 NSE stock futures (5-minute data, 2007–09), and the median basis is negative in Aggarwal (2015) and Nandan et al. (2014). The model reproduces this if reverse arbitrage is costly and cash-and-carry capital accepts a below-benchmark return. One candidate is the tax treatment of arbitrage funds, which are taxed as equity. If so, the inflows after the 1 April 2023 debt-fund tax change should have lowered r_A and with it the upper edge, which is a testable implication. It must be separated from the futures STT increase on the same date, which raises τ_+.

C.4. INVENTORY AND THE MOVING NO-TRADE REGION

The ban acts on existing positions, so the model needs inventory. Time is discrete (days). The arbitrage sector enters day t with inventory $a_{t-1} \geq 0$ and pays proportional costs on changes: c_+ per unit added and c_u per unit unwound (selling cash, buying back futures). To keep the dynamic analysis transparent, reverse arbitrage is set aside ($a_t \geq 0$): the capacity of stock *holders* to sell cash and buy futures when futures are cheap is part of the cash-market depth ψ_S. Each day the sector chooses a_t to maximise

$$a_t m_t - \tfrac{\gamma}{2} a_t^2 - c_+(a_t - a_{t-1})^+ - c_u(a_{t-1} - a_t)^+. \tag{C.7}$$

Proposition C.3 (Inventory-dependent no-trade region)**.** *Given a_{t-1}, the sector trades only when the basis leaves*

$$\left[\gamma a_{t-1} - c_u,\ \ \gamma a_{t-1} + c_+\right]. \tag{C.8}$$

In equilibrium, with $m_t^0 \equiv (u_t - a_{t-1})/\Phi$:

(a) if $m_t^0 > \gamma a_{t-1} + c_+$ the sector adds: $m_t = (\gamma u_t + c_+)/(\gamma\Phi + 1)$;

(b) if $m_t^0 < \gamma a_{t-1} - c_u$ it unwinds: $a_t = \max\{0, (u_t + \Phi c_u)/(\gamma\Phi + 1)\}$ and $m_t = (u_t - a_t)/\Phi$;

(c) otherwise $a_t = a_{t-1}$ *and* $m_t = m_t^0$.

Proof. (C.7) is strictly concave with kinks at a_{t-1}. The marginal value of the position is $m_t - \gamma a_t$, so adding is optimal iff $m_t - \gamma a_{t-1} > c_+$ and unwinding iff $m_t - \gamma a_{t-1} < -c_u$; interior solutions are $a_t = (m_t - c_+)/\gamma$ and $a_t = (m_t + c_u)/\gamma$. Uniqueness and the closed forms follow as in Proposition C.1, combining each branch with (C.3). □

Empirical content. Thresholds are not constants: they rise with arbitrage inventory. A large existing cash-and-carry book (proxied by open interest built while the basis was rich) raises both the level at which new arbitrage starts and the level at which unwinding starts. Threshold models should allow thresholds to depend on lagged OI.

C.5. THE F&O BAN AS A REDUCE-ONLY RULE

C.5.1 The rule

Once market-wide open interest in a stock's derivatives exceeds 95% of its MWPL at the end of a day, the stock enters the ban from the next day. Every entity may then only trade to reduce its position (from 1 October 2025, its end-of-day futures-equivalent delta); a change of sign counts as a violation. Normal trading resumes when open interest falls to 80% of MWPL (NSE Clearing circular 119/2025). The rule covers all expiries and options; rolling a position into the next expiry is not permitted. It binds outside traders as much as arbitrageurs.

Assumption C.2 (Ban)**.** During a ban:

(B1) *Reduce-only arbitrage:* $0 \le a_t \le a_{t-1}$.

(B2) *Blocked new exposure:* outside traders' demand shocks have two parts, $\Delta u_t = \varepsilon_t^n + \varepsilon_t^c$ in normal times. Closing demand ε_t^c (long liquidation, short covering) can still be executed and enters as usual. Demand for *new* futures exposure ε_t^n cannot; a share $\chi \in [0, 1]$ of it moves to the cash market and the rest is withdrawn. In the units of (C.3) the ban maps ε_t^n into $-\lambda\varepsilon_t^n$, with $\lambda \equiv \chi\,\psi_F/\psi_S \ge 0$.

(B2) follows from (C.3): new long demand $\Delta D_F = \varepsilon$ raises u by $\Phi\varepsilon/\psi_F$ if it enters futures, but lowers u by $\Phi\chi\varepsilon/\psi_S$ if a share χ of it buys cash instead. A leveraged retail trader who cannot buy the future and buys the stock on margin is the leading case.

Lemma C.4. *For the same* u_t *and* a_{t-1}*, the basis in a ban is at least as high as in normal times.*

Proof. (B1) removes only the option to add, which is used only when it lowers m (branch (a) of Proposition C.3); branches (b)–(c) are unchanged. □

Proposition C.5 (One-sided correction)**.** *Let a ban start with inventory* $a_{t-1} > 0$.

(i) No ceiling. *For $u_t \geq a_{t-1}(1+\gamma\Phi) - \Phi c_u$ the basis is $m_t = (u_t - a_{t-1})/\Phi$: every increase in pressure passes through fully ($\partial m/\partial u = 1/\Phi$ instead of $\gamma/(\gamma\Phi+1)$ beyond the normal upper edge). Rich futures are not corrected.*

(ii) A floor while inventory lasts. *For lower u_t the sector unwinds and $m_t = \gamma a_t - c_u$, so cheap futures are still corrected, at a level that* rises *with the remaining inventory.*

(iii) The floor wears out. *Once $a_t = 0$ the floor disappears and $m_t = u_t/\Phi$. Because positions cannot be rolled, an expiry inside a ban removes the expiring part of the inventory for the rest of the ban.*

Proof. Apply Proposition C.3 with branch (a) deleted by (B1). Branch (c) holds while $m_t^0 \geq \gamma a_{t-1} - c_u$, i.e. $u_t \geq a_{t-1}(1+\gamma\Phi) - \Phi c_u$, giving (i). In branch (b), $a_t = (m_t + c_u)/\gamma$ when interior, so $m_t = \gamma a_t - c_u$, giving (ii). If $(u_t + \Phi c_u) \leq 0$ then $a_t = 0$ and $m_t = u_t/\Phi$, giving (iii). Expiry: delivered positions close and cannot be re-opened in the next expiry under the ban. □

Proposition C.6 (Sign reversal of new-exposure demand)**.** *In normal times a shock to demand for new futures exposure moves the basis in its own direction: $\partial m_t/\partial \varepsilon_t^n = 1/\Phi$ inside the no-trade region and $\gamma/(\gamma\Phi+1)$ beyond its upper edge. In a ban the same shock moves it the other way: $\partial m_t/\partial \varepsilon_t^n = -\lambda/\Phi$ inside the no-trade region and $-\lambda\gamma/(\gamma\Phi+1)$ on the unwinding floor. The reversal is strict iff $\chi > 0$.*

Proof. Chain rule on Propositions C.3–C.5, with $\partial u/\partial \varepsilon^n = 1$ in normal times and $-\lambda$ in a ban by (B2). □

Proposition C.7 (The level effect is ambiguous)**.** *Fix a_{t-1}, the pre-shock pressure u^0 and closing demand ε^c, and compare the same new-exposure shock $\varepsilon^n \geq 0$ with and without a ban: $m^N = m(u^0 + \varepsilon^c + \varepsilon^n)$ and $m^B = m^{ban}(u^0 + \varepsilon^c - \lambda\varepsilon^n)$. Then:*

(i) *$m^B - m^N$ is non-increasing in ε^n;*

(ii) *at $\varepsilon^n = 0$, $m^B - m^N \geq 0$, strictly if closing pressure would have triggered new arbitrage in normal times;*

(iii) *$m^B - m^N < 0$ for ε^n large enough.*

The ban raises *the basis when pressure comes mainly from closing trades (e.g. short covering) and* lowers *it when new speculative demand keeps arriving; the observable sign condition is the composition of order flow during the ban.*

Proof. m^N is non-decreasing in ε^n; m^B is non-increasing in ε^n because u^B falls in ε^n and m^{ban} is non-decreasing in u. Hence (i). (ii) is Lemma C.4. (iii): m^N rises without bound (slope at least $\gamma/(\gamma\Phi+1) > 0$) while m^B is non-increasing. □

Proposition C.8 (Trading and measurement in a ban)**.** *(i) If the reduce-only rule binds contract by contract, every futures trade in a ban closes a long and a short, so $\Delta OI_t = -V_t$ (volume) outside expiry days. Trades that offset across instruments (e.g. delta-reducing option trades allowed from 1 October 2025) break the identity; the gap measures such offsetting.*

(ii) Trading requires that some long and some short both want to reduce at the same price, so futures volume falls and quoted spreads widen relative to normal days.

(iii) On days without trades in the last half hour, NSE sets the futures settlement price at a theoretical cost-of-carry value. The settlement-price basis is then zero by construction (up to differences between NSE's rate and r_b).

Proof. (i) A buyer may only reduce a short and a seller only reduce a long, so each trade lowers OI by its size. (ii) follows from the restriction of the trading set. (iii) is the NSE settlement rule. □

Remark. Part (iii) is a trap for the empirics. Bans thin out trading, so a naive basis built from settlement prices would show the basis "closing" during bans for purely mechanical reasons. The data work must use trade-based prices and flag theoretical settlements.

C.6. ANTICIPATION AND HYSTERESIS

Assumption C.3 (Exit in a ban is costlier)**.** If a ban occurs before a position is closed, unwinding costs $c_u + \ell$ with $\ell > 0$, because the counterparties are only traders who also want to reduce (Prop. C.8(ii)). The probability that a position opened when utilisation is $\omega = OI/MWPL$ meets a ban before it is closed is $\pi(\omega)$, continuous and increasing.

Proposition C.9 (Anticipation)**.** *Outside a ban the effective cost of opening cash-and-carry is $c_+ + \pi(\omega)\ell$. The upper edge of the no-trade region rises smoothly as utilisation approaches 95%, and new arbitrage slows before any ban. The jump at $\omega = 0.95$ consists only of the reduce-only effects of Section C.5; anticipation is continuous through the threshold.*

Proof. The expected cost of a unit opened at ω is $c_+ + \pi(\omega)\ell$ (exit at a later date is costlier with probability π); substitute for c_+ in Proposition C.3. Continuity of π gives continuity at 0.95. □

Remark (What the density of utilisation should look like)*.* Arbitrageurs hold back as ω rises, which thins the density just below 95%; speculators who expect a ban may rush to open positions first, which does the opposite. The model therefore does not sign a McCrary test; the test tells us which force dominates.

Corollary C.10 (Hysteresis)**.** *For $\omega \in [0.80, 0.95)$ a stock may be in a ban (it crossed 95% earlier) or not. At the same ω the first is governed by Section C.5, the second by Proposition C.9. Their difference is the effect of the reduce-only state relative to the anticipation state, which is the effect the 95/80 rule creates.*

C.7. WHAT THE MODEL IMPLIES FOR IDENTIFICATION

Proposition C.11 (MWPL is not a clean instrument for ban status)**.** *The basis depends on the MWPL through three channels: the ban state $B_t = B(\omega_t, B_{t-1})$, the anticipation term $\pi(\omega_t)$, and*

entity limits proportional to the MWPL when they bind. A change in the MWPL therefore moves the basis even in periods with no ban, so it violates the exclusion restriction for an instrument of B_t. Denominator variation identifies the total *effect of the position-limit regime (ban plus anticipation plus caps), not the effect of the ban alone.*

Proof. $\omega = OI/MWPL$ enters $\pi(\omega)$ (Prop. C.9) independently of B. □

What each design identifies under the model:

1. *RD at 95%*: the local jump in the reduce-only effects, provided utilisation cannot be placed precisely on one side.
2. *Hysteresis comparison* (Cor. C.10): the reduce-only state against the anticipation state at equal ω; selection on past paths must be controlled (lagged utilisation, lagged basis).
3. *Ban-entry event study with near-miss controls*: Props. C.5–C.8 dynamically, against stocks with similar paths that stayed below 95%.
4. *1 October 2025 reform and quarterly MWPL recalculations*: reduced-form effects of predicted changes in utilisation (new formula; FutEq numerator) on the basis. These are total-regime effects (Prop. C.11).

C.8. SHORT-SALE CONSTRAINTS BEYOND THE BASIS

C.8.1 Earnings news in a ban

Proposition C.12 (Bad news is incorporated more slowly in a ban)**.** *Let an announcement reveal news $\pm\delta$, and let trading on the day impound a share k of it. For bad news, a share ρ of informed selling capacity runs through short futures, the only overnight short in an F&O stock. In a ban that capacity is unavailable, so the share impounded falls to $k(1-\rho)$ and the post-announcement drift after bad news rises from $(1-k)\delta$ to $(1-k+k\rho)\delta$. Good news is unaffected because optimists can buy cash.*

Proof. Immediate from the definitions; the asymmetry comes from the fact that the cash market allows buying but not overnight selling by non-holders. □

Remark. Outside a ban the same mechanism predicts that informed pessimists push the futures price *below* carry before bad news, since their short futures cannot be met by reverse arbitrage. A recent working paper (Saurav, Srivastava and Agarwalla, 2026) reports a negative basis before earnings announcements in Indian stock futures. The model adds that this pre-announcement signal should *vanish* for stocks in a ban, while drift after bad news should grow.

C.8.2 Cash-only stocks and F&O inclusion

Stocks outside the F&O segment have no single-stock futures and are not eligible for short selling under SEBI's framework; stock lending, where it exists, is thin. So c_- is effectively infinite

for overnight positions. A Miller (1977) mechanism then applies.

Proposition C.13 (Overpricing without a short channel)**.** *A continuum of investors has valuations $v_i \sim U[v-\delta, v+\delta]$ and demands $k(v_i - p)$ if long; shorting costs c_s per unit, so investors with $v_i < p - c_s$ short $k(v_i - p + c_s)$. Supply is $Q > 0$.*

(i) The price $p(c_s)$ is non-decreasing in c_s.

(ii) With free shorting $p = v - Q/k$. With no shorting and $\delta \geq Q/k$, $p = v + \delta - 2\sqrt{\delta Q/k}$, so the overpricing is $\left(\sqrt{\delta} - \sqrt{Q/k}\right)^2$, increasing in dispersion δ.

Proof. (i) Aggregate demand falls in p and rises in c_s (short positions shrink), so the market-clearing price rises in c_s. (ii) Free shorting: $\int k(v_i - p)\, di = k(v-p) = Q$. No shorting: $\frac{k}{2\delta}\int_p^{v+\delta}(x-p)\,dx = \frac{k(v+\delta-p)^2}{4\delta} = Q$, valid when $p \geq v - \delta$, i.e. $\delta \geq Q/k$. Subtract; the derivative in δ is $1 - \sqrt{Q/(k\delta)} > 0$. □

Empirical content. F&O inclusion lowers c_s (futures, options and short-selling eligibility arrive together). It should lower prices of high-dispersion stocks at inclusion and reduce their subsequent underperformance; exclusion should do the reverse. Diamond and Verrecchia (1987) imply that, even without a level bias, short constraints slow the incorporation of bad news, which points in the same direction as Proposition C.12.

C.9. HETEROGENEOUS ARBITRAGEURS AND THE ESTIMATION LENS

With a single cost c_+ the band has sharp edges and, under impulse-style correction, the outer regimes would almost never be observed. Let costs differ across arbitrageurs, $c \sim G$ on $[\underline{c}, \bar{c}]$, each with the objective (C.4). Aggregate arbitrage is $a(m) = \int (m-c)^+\, dG(c)/\gamma$, which is continuous and convex in m above $\underline{c}$, so the strength of correction rises smoothly with the size of the deviation. In discrete time, with capital moving to opportunities only gradually, the basis follows an error-correction process whose speed of adjustment is a smooth, increasing function of $|m_{t-1} - \text{centre}|$. This is the exponential or logistic smooth-transition model (Tse, 2001). It is also well approximated by a three-regime threshold model whose thresholds are quantiles of G (Dwyer, Locke and Yu, 1996; Ters and Urban, 2020). Proposition C.3 adds that the thresholds move with inventory, and Proposition C.5 that in a ban the upper regime's adjustment speed falls toward zero while the lower regime's remains. A band-of-inaction (impulse-control) derivation gives the same reading for a single arbitrageur: the threshold is the point at which correcting the deviation becomes worth its cost.

C.10. CALIBRATION AND HORIZON

For an overnight carrier, the round-trip cost of a cash-and-carry trade includes STT on the cash purchase and sale (0.1% each), STT on the futures sale (0.01–0.05% depending on the period) or

0.1% on physical delivery, stamp duty, exchange and clearing fees, brokerage and impact. That is tens of basis points, far wider than the bands of a few basis points estimated from one-second data (Aggarwal, 2015), which describe intraday market makers. Arbitrage funds that roll rather than deliver spread these costs over several expiries, which lowers the effective τ_+ per month. Chapter 6 computes a bottom-up cost band for each period; the simulations here use round illustrative numbers ($c_+ = 15$, $c_u = 10$, $c_- = 60$ bp) and are not estimates.

C.11. FROM PROPOSITIONS TO HYPOTHESES

H	SOURCE	PREDICTION	WHAT WOULD CONTRADICT IT
H1	Props. C.1–C.3	Mispricing size and persistence rise with ex-ante frictions	No relation between friction index and band/persistence
H2	Props. C.12, C.13	Drift after bad news larger without a short channel (cash-only stocks; ban days); good news unaffected	Symmetric drift response, or larger effect for good news
H3	Prop. C.2(i),(iii)	Edges move one-for-one with cost changes (STT 2023, 2024, 2026); edges shrink toward expiry; estimated band close to cost band	Band unrelated to dated cost changes
H4	Prop. C.2(ii),(iv)	Scarcer capital widens excursions beyond the edges; tax-advantaged inflows lower the upper edge	Basis level unaffected by arbitrage-fund flows (after separating the 1 Apr 2023 STT change)
H5	Prop. C.13	Overpricing-side anomalies stronger where shorting is costly; underpricing side unaffected	Symmetric effect on both legs
H6	Props. C.12, C.13	F&O inclusion lowers overpricing and speeds the incorporation of bad news (tested through price delay and drift after bad news); exclusion does the reverse	No change, or opposite sign
H7	Prop. C.5	In a ban, positive deviations stop reverting; negative ones still revert while inventory lasts, less after an expiry in the ban	Symmetric change, or no change
H8	Props. C.6, C.8	In a ban, futures volume falls, $\Delta OI \approx -V$, cash activity rises; response of the basis to new-demand shocks flips sign	Futures activity unchanged; no sign flip
H9	Props. C.9, C.11, Cor. C.10	Upper edge rises smoothly as utilisation nears 95%; the 1 Oct 2025 package acts through predicted utilisation, with sign set per stock	Jump only at 95% with no pre-trend; no effect of predicted utilisation

The level effect of the ban (Prop. C.7) is deliberately *not* a one-sided hypothesis: the test is whether its sign follows the composition of order flow (closing versus new demand) as the model says.

C.12. LIMITATIONS

1. Partial equilibrium with linear outside demand and a myopic, mean–variance arbitrage sector; dynamic hedging and the option to unwind early (Brennan and Schwartz, 1990) enter only through the inventory structure.
2. The split of outside demand into closing and new-exposure parts (B2) is the modelling choice that drives Props. C.6–C.7. Its empirical counterpart (order flow that opens versus closes positions) is only partly observable from public data (changes in OI against volume).
3. Reverse arbitrage by stock holders is folded into cash-market depth, not modelled as a separate agent.
4. Anticipation rests on Assumption C.3 (costlier exit in a ban), which can itself be tested (Proposition C.8(ii)).
5. Propositions C.12–C.13 are deliberately simple; they fix directions, not magnitudes.

C.13. NUMERICAL VERIFICATION AND ILLUSTRATION

A script solves each equilibrium by brute force (bisection on the best-response fixed point) and compares it with the closed forms. All checks pass: the static band (Prop. C.1), the inventory step in normal times and in a ban (Props. C.3–C.5), the floor $\gamma a_t - c_u$ and the full passthrough above it, Lemma C.4, the monotonicity and sign change in Proposition C.7, and the Miller closed forms (Prop. C.13).

The simulation uses illustrative parameters ($\Phi = \gamma = 1$; $c_+ = 15$, $c_u = 10$ bp; demand pressure AR(1) with persistence 0.9 around 25 bp; daily shock s.d. 6 bp for each component; $\lambda = 0.5$). Each experiment runs the same shocks with and without a 40-day ban.

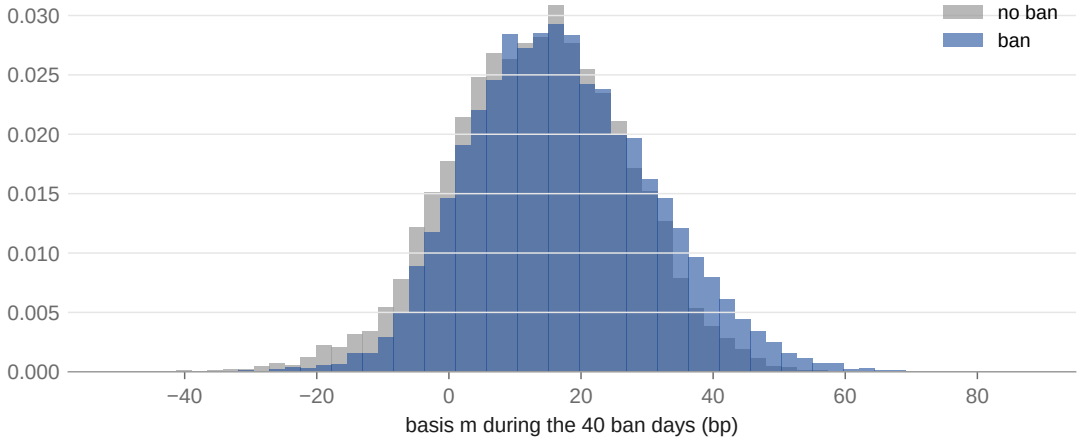
- *Asymmetric reversion* (300 episodes): the slope of Δm_t on m_{t-1} above the 80th percentile weakens from -0.18 (no ban) to -0.13 (ban); below the 20th percentile it strengthens from -0.12 to -0.16.
- *Level effect*: with no extra new-exposure demand during the ban the mean basis is +4.9 bp higher; with a steady inflow of 1, 2 and 4 bp/day of new demand it is -3.1, -10.8 and -26.0 bp lower (Prop. C.7).

The static band and one simulated path are drawn in Figures 4.1 and 4.2 of Chapter 4.

FIGURE C.1

With the same shocks, simulated bans shift the distribution of the basis

Distribution of the basis over the 40 ban days, pooled across 300 paired simulations



Source: model simulations; illustrative parameters, not estimates.

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